
THE SHAPE OF ATTRACTION IN UMAP: EXPLORING THE EMBEDDING FORCES IN DIMENSIONALITY REDUCTION

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ABSTRACT

Uniform manifold approximation and projection (UMAP) is a widely used neighbor-embedding procedure for constructing low-dimensional representations. Its optimization objective induces attractive and repulsive update terms between data points, and the resulting embedding geometry is determined by the distance-dependent form of these terms. This paper analyzes these forces as functional objects, characterizes their effects on cluster formation and visualization, and compares UMAP with related embedding methods. Repulsion primarily encodes dissimilarity by regulating cluster boundaries and inter-cluster separation. Attraction has a more intricate role: the same attractive update can induce either distance contraction or distance expansion in the embedding space, depending on the current pairwise distance and learning-rate scale. This observation explains the operational need for learning-rate annealing and clarifies why attractive and repulsive terms are treated asymmetrically. We further show that an explicit modification of the attraction shape improves cluster-placement consistency under random initialization. Overall, the proposed analysis provides a more interpretable, robust, and technically controlled account of UMAP and related neighbor-embedding algorithms.

1 Introduction

Modern scientific and industrial pipelines routinely produce observations with high ambient dimension. Dimensionality reduction (DR) methods provide an exploratory analysis mechanism for such observations by constructing low-dimensional coordinates that expose salient geometric or topological structure. Prominent methods, including t -distributed stochastic neighbor embedding (Maaten and Hinton [2008]) and uniform manifold approximation and projection (UMAP) (McInnes et al. [2018]), implement this construction through attraction–repulsion dynamics: pairs judged similar in the input space are assigned update terms that reduce their embedding distance, whereas dissimilar or negatively sampled pairs receive update terms that increase separation. Because these procedures are unsupervised, they do not use label information during optimization; instead, they attempt to preserve intrinsic high-dimensional structure through local attractive relations and global repulsive relations. This mechanism makes them useful for clustering (Becht et al. [2019]), exploratory data analysis (Fleischer and Islam [2020]), anomaly detection in semiconductor manufacturing (Fan et al. [2021]), visual search (González-Márquez et al. [2024]), time-series analysis (Altin and Cakir [2024]), analysis of representation convergence (Huh et al. [2024]), and outlier image detection (Islam and Fleischer [2024]), where unlabeled data must be inspected through geometric organization. In this sense, DR instantiates a label-free representation procedure that converts complex high-dimensional measurements into visually interpretable coordinate systems.

Subsequent methods extend the same attraction–repulsion principle (Amid and Warmuth [2019]; Agrawal et al. [2021]; Wang et al. [2021]; Narayan et al. [2021]; Yang et al. [2022]; Wang et al. [2025]; Kury et al. [2025]), typically by changing which relations are emphasized or how pairwise interactions are weighted. Although these algorithms are empirically useful across many application domains, their practical use still depends on heuristic choices that do not always admit a precise interpretation. In addition, any low-dimensional embedding necessarily introduces distortion (Chari and Pachter [2023]). A more explicit account of the optimization dynamics is therefore required if practitioners are to distinguish meaningful embedding structure from algorithm-induced artifacts and to tune performance in a principled manner. Current implementations often obtain compact clusters through initialization choices, learning-rate schedules, and kernel-parameter adjustments, yet the distance-dependent attractive and repulsive mechanisms that produce these

clusters remain only partially characterized. Moreover, important tunable quantities are embedded inside abstract functional forms, which obscures their mechanistic role.

This paper decomposes the embedding updates into constituent force terms and studies the induced functional shapes of attraction and repulsion. The analysis identifies attraction as the term that determines the need for learning-rate annealing, the difficulty of obtaining consistent embeddings from random initialization, and the mechanism by which clusters form. Repulsion, in contrast, primarily controls inter-cluster separation once clusters are established. The specific contributions are as follows:

1. We derive attraction and repulsion shapes from the corresponding update forces, establish sufficient conditions for pairwise distance contraction and expansion, and use these conditions as an analytic lens for neighbor-embedding algorithms (Section 4).
2. We show that the UMAP attraction shape can produce both distance contraction and distance expansion, and we use cross-method comparisons to specify how attraction constrains the learning-rate annealing scheme (Section 5).
3. We introduce a modified attraction shape that improves embedding consistency under random initialization, indicating that the optimization can recover a reproducible latent structure when long-range attraction is strengthened (Section 5.1).
4. We analyze repulsion shapes to characterize their role in cluster compactness and inter-cluster-distance regulation (Section 5.2).
5. We compare the attraction and repulsion shapes of UMAP, Parametric UMAP, and NEG- t -SNE, identify their formal similarities and differences, and relate these shapes to algorithmic stability (Section 6).

The main text focuses on UMAP because it is now a de facto standard in many application areas; other algorithms are introduced where their force shapes clarify the analysis. Additional algorithmic details are provided in the appendix. The code used in this study will be released upon acceptance.

2 Related Works

Modern iterative graph-based neighbor-embedding algorithms originate with stochastic neighbor embedding (SNE) (Hinton and Roweis 2002) and its t -distribution extension, t -SNE (Maaten and Hinton 2008). These methods construct dense pairwise graphs in which each datum is related to every other datum, including pairs that are not close in the input space. The associated graph weights are normalized so that they can be interpreted as probability distributions. Concurrent spectral methods, including locally linear embedding (Roweis and Saul 2000) and Laplacian Eigenmaps (LE) (Belkin and Niyogi 2002), instead operate on k -nearest-neighbor (k -NN) graphs of pairwise interactions and solve eigenvalue-decomposition problems. Tang et al. (Tang et al. 2016) subsequently inserted the k -NN graph into an iterative optimization framework and removed the low-dimensional normalization. UMAP further extends this formulation by eliminating normalization in both the high- and low-dimensional spaces and by optimizing only pairwise interaction terms (McInnes et al. 2018). Its update rule uses an explicit attractive term to preserve local neighborhoods and a repulsive term to separate dissimilar points. Methods such as PaCMAP (Wang et al. 2021) and NEG- t -SNE (Damrich et al. 2023) build on this family of formulations. For a recent survey of low-dimensional embedding methods, see de Bodt et al. (2025).

Substantial prior work has analyzed the formal relationships among these algorithms. An early analysis of SNE showed that well-clustered data in the original space remain well-clustered in the embedding space (Shaham and Steinerberger 2017). For t -SNE, Linderman and Steinerberger (2019) proved that the number of embedded clusters provides a lower bound on the number of clusters in the original space, and subsequent studies refined this characterization (Arora et al. 2018; Cai and Ma 2022; Linderman and Steinerberger 2022). Because t -SNE and UMAP arise from related neighbor-embedding principles but can produce markedly different visual geometries, several studies have sought a unified account (Böhm et al. 2022; Damrich et al. 2023; Draganov et al. 2023). Böhm et al. (2022) proposed that Laplacian Eigenmaps, ForceAtlas2 (Jacomy et al. 2014), UMAP, and t -SNE occupy positions on a common algorithmic spectrum. LE and t -SNE are linked through the early-exaggeration phase (Cai and Ma 2022), and UMAP can be related to t -SNE through contrastive estimation (Damrich et al. 2023). Independently, Hu et al. (2023) identified a connection between contrastive learning and SNE. Other recent explanations use probabilistic formulations (Ravuri et al. 2023; Ravuri and Lawrence 2024), kernel-based analyses (Draganov and Dohn 2023), and information geometry (Kolpakov and Rocke 2024).

3 Uniform Manifold Approximation and Projection

UMAP constructs a high-dimensional graph over the dataset $X = \{x_i \in \mathbb{R}^n | i = 1, \dots, N\}$ by assigning pairwise weights $p_{i,j} = f_h(d(x_i, x_j))$, typically with values in $[0, 1]$ (for details, see Appendix H).

The corresponding graph in the low-dimensional representation $Y = \{y_i \in \mathbb{R}^d | i = 1, \dots, N\}$ is parameterized by the differentiable affinity function

$$q_{ij} = \frac{1}{1 + a(\|y_i - y_j\|_2^2)^b}, \quad (1)$$

where the parameters a and b determine the density of the mapping and are chosen by fitting q_{ij} to

$$\Psi(d_{ij}) = \begin{cases} 1 & \text{if } \|y_i - y_j\|_2 < m_d \\ \exp(-(d_{ij} - m_d)) & \text{otherwise} \end{cases}, \quad (2)$$

where m_d specifies the target separation between the two nearest low-dimensional points.

UMAP then minimizes the following cross-entropy objective:

$$\mathcal{L} = \sum_{i,j} (-p_{ij} \log(q_{ij}) - (1 - p_{ij}) \log(1 - q_{ij})). \quad (3)$$

The first term induces attraction, whereas the second term induces repulsion. Rather than optimizing over all pairs at each iteration, UMAP uses negative sampling (Mikolov et al. [2013], Tang et al. [2016]). For every edge satisfying $p_{ij} > 0$, referred to as a positive edge, the algorithm samples several negative edges at random. The attractive update is applied to the positive edge:

$$y_i^{t+1} = y_i^t + \lambda \nabla_{y_i^t} \log(q_{ij}), \quad (4)$$

$$y_j^{t+1} = y_j^t + \lambda \nabla_{y_j^t} \log(q_{ij}), \quad (5)$$

and the repulsive update is applied to the negative edges:

$$y_i^{t+1} = y_i^t + \lambda \nabla_{y_i^t} \log(1 - q_{ij}), \quad (6)$$

where $\lambda(> 0)$ is the learning rate and t is the step number. Note that y_j is not updated for negative edges. For a detailed analysis of UMAP's loss, see Damrich and Hamprecht (2021).

4 Attraction and Repulsion Shapes

The updates in (4-6) can be expressed by decomposing the gradients $\nabla_{y_i^t} \log(q_{ij})$ and $\nabla_{y_j^t} \log(1 - q_{ij})$ into a distance-dependent scalar coefficient multiplying the displacement vector $(y_i - y_j)$. We refer to this scalar coefficient as the attraction shape or repulsion shape, depending on the loss term from which it is derived. Although UMAP is the running example, the same decomposition applies to any embedding method with attraction–repulsion updates.

Writing $\nabla_{y_i^t} \log(q_{ij}) = f_a(\zeta^t)(y_i^t - y_j^t)$, with $\zeta = \|y_i - y_j\|_2$, yields the following positive-edge updates:

$$y_i^{t+1} = y_i^t + \lambda f_a(\zeta^t)(y_i^t - y_j^t), \quad (7)$$

$$y_j^{t+1} = y_j^t - \lambda f_a(\zeta^t)(y_i^t - y_j^t). \quad (8)$$

Here, $f_a : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}_{\leq 0}$ is the attraction shape, and the Euclidean metric gives $\nabla_{y_i^t} \log(q_{ij}) = -\nabla_{y_j^t} \log(q_{ij})$. The negative-edge update can be written analogously as

$$y_i^{t+1} = y_i^t + \lambda f_r(\zeta^t)(y_i^t - y_j^t), \quad (9)$$

$$y_j^{t+1} = y_j^t, \quad (10)$$

where $f_r : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}_{\geq 0}$ is the repulsion shape.

Related decompositions appear in prior work (McInnes et al. [2018], Agrawal et al. [2021], Draganov and Dohn [2023]), although their definitions and uses differ. The original UMAP paper (McInnes et al. [2018]) used the decomposition primarily as a computational device for efficient optimization, whereas Draganov and Dohn (2023) used derivative forms to compare algorithms from a kernel perspective. In both settings, the derivative computation itself is the central object. Here, we instead treat the decomposed coefficients as independent distance-indexed functions that can be analyzed and modified. A related representation was used by Agrawal et al. (2021), who expressed the update as a scalar coefficient multiplied by the unit direction $((y_i - y_j)/\|y_i - y_j\|_2)$ in order to emphasize force magnitude. The analysis below shows that the full shape of the scalar function is more informative than magnitude alone.

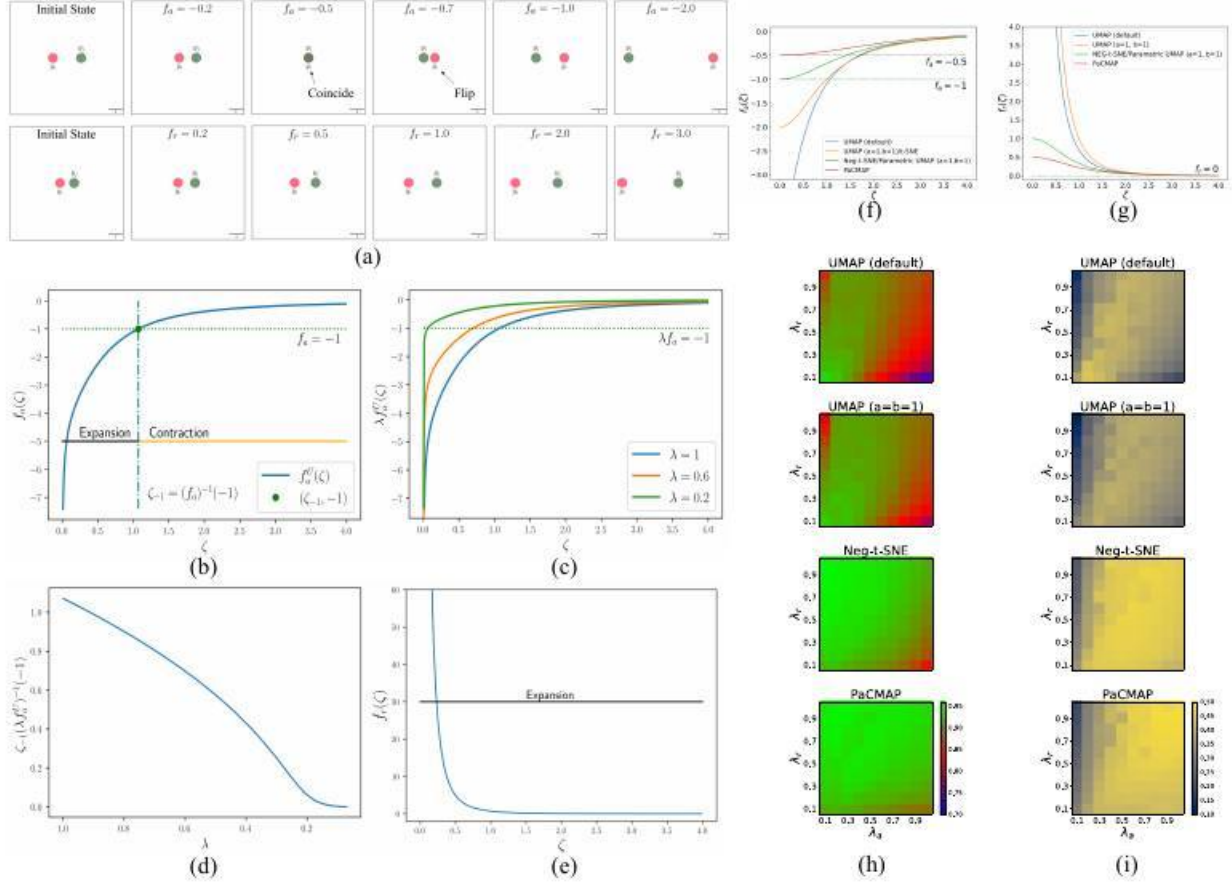


Figure 1: Attraction and repulsion shapes in UMAP. (a) Pairwise effect of selected values of f_a (top) and f_r (bottom). (b) UMAP attraction shape. (c) Effective attraction shape (λf_a) under multiple learning rates λ . (d) Minimum contraction distance (ζ_{-1}) as λ decreases. (e) UMAP repulsion shape. (f) Attraction and (g) repulsion shapes for representative embedding methods. (h) Trustworthiness and (i) silhouette score obtained when the constant attraction learning rate (λ_a) and repulsion learning rate (λ_r) are varied independently. Default UMAP parameters: $a = 1.58$ and $b = 0.89$.

4.1 Conditions for Attraction and Repulsion

This section derives attraction and repulsion conditions directly from the update equations (7)-(10). The following proposition gives a sufficient condition for contraction of the distance between the positive-edge pair y_i^t and y_j^t :

Proposition 4.1. *The update Eqs. (7) and (8) provide a contraction of distance ($\|y_i^{t+1} - y_j^{t+1}\| < \|y_i^t - y_j^t\|$) if $-1 < \lambda f_a < 0$.*

Here, λf_a is the effective attraction shape. In most analyses, however, the unscaled function f_a is sufficient for interpreting embedding behavior. For a negative edge, the following proposition gives a sufficient condition for expansion of the distance between y_i^t and y_j^t :

Proposition 4.2. *The update Eqs. (9) and (10) provide an expansion of distance ($\|y_i^{t+1} - y_j^{t+1}\| > \|y_i^t - y_j^t\|$) if $f_r > 0$.*

Adding a symmetric term $-\lambda f_r(\zeta)(y_i^t - y_j^t)$ to Eq. (10) does not change this conclusion. These conditions should be interpreted as per-iteration certificates for guaranteed contraction or expansion, not as statements about learning-rate tuning or decay. In particular, the shapes can be designed so that contraction and expansion properties do not depend on learning-rate annealing.

Figure 1(a) illustrates how selected values of $f_a < 0$ and $f_r > 0$ affect a two-point configuration. The repulsion condition is direct: any positive value of f_r expands the pairwise distance. The attraction coefficient is less direct because f_a can encode either contraction or expansion. For $f_a \in (-1.0, 0)$, the update decreases the distance, with

maximum contraction occurring at $f_a = -0.5$ when the points coincide after the update. Values below -1.0 instead increase the distance. Thus, although the update is local at the level of a sampled pair, the distance-dependent shape of these local terms determines the emergent global geometry.

5 Analysis of UMAP in terms of Attraction and Repulsion Shape

Using the gradient decomposition and the distance form [1](#), the attraction and repulsion shapes are given by

$$f_a^U(\zeta) = -\frac{2ab\zeta^{2(b-1)}}{1 + a\zeta^{2b}} \quad (11)$$

and

$$f_r^U(\zeta) = \frac{2b}{\zeta^{2b}(1 + a\zeta^{2b})}, \quad (12)$$

respectively. This section analyzes the default UMAP shapes, which are controlled mainly by the parameters a and b , and interprets the consequences of perturbing these parameters. Unless otherwise noted, the discussion applies to $b \leq 1$, where the attraction shape is strictly increasing and therefore invertible (for derivation, see [Appendix A](#)).

[Figure 1\(b\)](#) displays the default UMAP attraction shape ($a = 1.58, b = 0.89$). The function is unbounded and approaches ∞ in magnitude as $\zeta \rightarrow 0$. By [Proposition 4.1](#), the update changes from contraction to expansion when λf_a^U crosses -1 as ζ approaches 0. Because the attraction shape is invertible in the parameter regime considered here, the transition distance $\zeta_{-1} = (\lambda f_a^U)^{-1}(-1)$ can be interpreted as the minimum distance at which an attractive update still contracts the pair. Equivalently, $\zeta > \zeta_{-1}$ yields contraction, whereas $0 < \zeta < \zeta_{-1}$ yields expansion. This behavior contradicts the informal interpretation that an attractive update always moves neighboring points closer together.

When ζ_{-1} is large, neighboring points can alternate between contraction and expansion, producing visually diffuse cluster boundaries. Sharper cluster boundaries therefore correspond to the limiting regime $\zeta_{-1} \rightarrow 0$. UMAP achieves this regime through a learning-rate schedule that anneals toward zero ([Figs. 1\(c,d\)](#)). In contrast, the repulsion shape in [Eq. 12](#) is positive for its domain and therefore satisfies [Proposition 4.2](#). The function f_r^U approaches 0 as $\zeta \rightarrow \infty$ and diverges as $\zeta \rightarrow 0$ ([Fig. 1\(e\)](#)).

Comparing attraction shapes across algorithms shows that UMAP and t-SNE are the methods for which $f_a < -1$ can occur (and therefore $\zeta_1 > 0$ when $\lambda = 1$; [Fig. 1\(f\)](#)). t-SNE mitigates this regime by weighting updates with the corresponding p_{ij} values, whereas UMAP relies on learning-rate annealing. Neg-t-SNE and PaCMAP have attraction shapes that lie naturally in $[-1, 0]$ and therefore satisfy [Proposition 4.1](#) for $\lambda \in [0, 1]$ with $\zeta_{-1} = 0$. In addition, PaCMAP’s weighted f_a is always less than 0.5 in magnitude, which prevents sign-flip behavior during attractive updates. The repulsion-shape comparison shows that UMAP is the method with large values at small distances ([Fig. 1\(g\)](#)); the shape is unbounded, although implementations often clip the values during optimization. Additional details for these and related methods are provided in [Appendix G](#).

We evaluate these attraction and repulsion choices on MNIST by assigning separate constant learning rates to attraction (λ_a) and repulsion (λ_r) and sweeping their values ([Figs. 1\(h,i\)](#)). For UMAP, smaller λ_a values (< 0.5 , and consequently $\zeta_{-1} \simeq 0$) produce better embeddings as measured by trustworthiness (T) ([Venna and Kaski 2001](#)), a local-structure-preservation metric ([Fig. 1\(h\)](#)), and better clustering as measured by silhouette score (SIL) ([Rousseeuw 1987](#)), a cluster-separation metric ([Fig. 1\(i\)](#)). By contrast, the full range $\lambda_r \in [0, 1]$ remains usable. For NEG-t-SNE and PaCMAP, where $\zeta_{-1} = 0$, the complete parameter range is effective. These sweeps confirm that UMAP requires ζ_{-1} to be close to 0 for high-quality embedding and clustering, and that the standard implementation obtains this condition through learning-rate annealing; newer algorithms encode $\zeta_{-1} = 0$ by default. Additional results and illustrations on other datasets are provided in [Section 6](#) and [Appendices B, G](#) and [I](#).

5.1 Improving UMAP’s Consistency under Random Initialization

UMAP embedding consistency depends strongly on initialization ([Kobak and Linderman 2021](#), [Wang et al. 2021](#)). In standard workflows, the initial embedding is obtained either from principal component analysis (PCA) of the data or from a spectral decomposition of the high-dimensional graph, and these initializations produce relatively consistent mappings despite stochastic optimization components. Under purely random initialization, however, clusters may fail to form or may appear in a different orientation in each run. When two high-dimensional nearest neighbors are initialized far apart in the embedding space, the corresponding attractive update can be too weak to contract the pair effectively. This near-sightedness phenomenon ([Wang et al. 2021](#)) is encoded directly in the attraction shape, because $|f_a^U|$ decreases toward zero as distance increases (i.e., $\lim_{\zeta \rightarrow \infty} |f_a^U(\zeta)| = 0$).

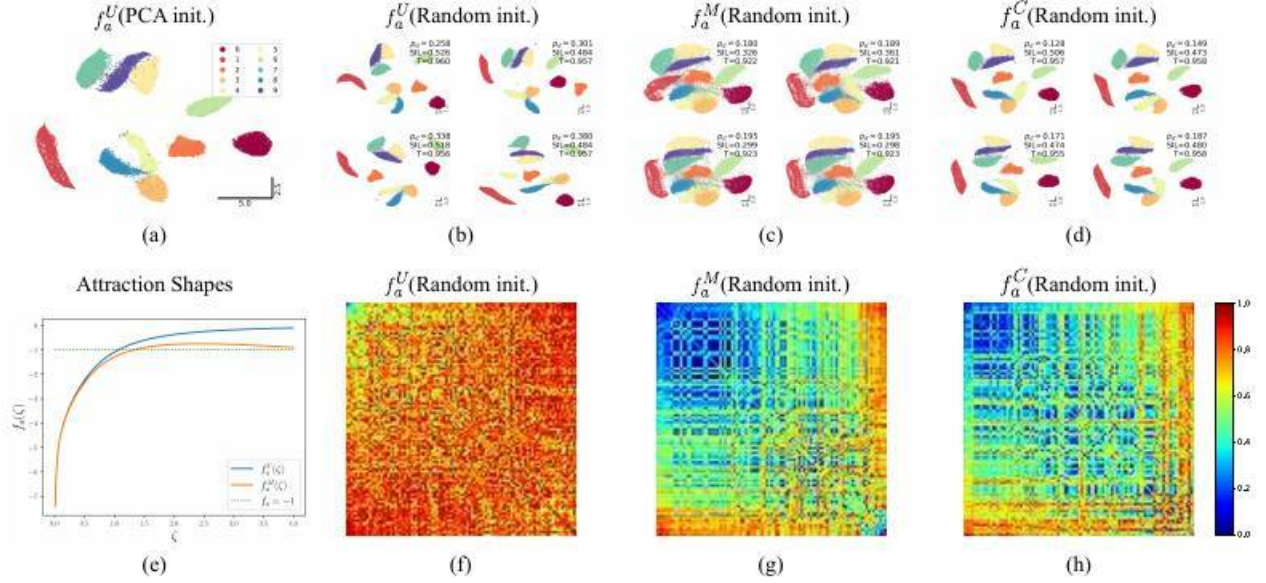


Figure 2: Effect of random initialization on MNIST embeddings under different attraction shapes. (a) PCA-initialized reference mapping. (b-d) Four mappings with the smallest Procrustes distance (p_d) from the reference in (a), shown for (b) UMAP, (c) the modified attraction shape, and (d) the composite attraction shape. (e) Default UMAP and modified attraction shapes. (f-h) Procrustes matrices computed from 100 runs for (f) UMAP (0.78 ± 0.13), (g) the modified shape (0.49 ± 0.21), and (h) the composite shape (0.50 ± 0.20). Diagonal (i, i) entries are sorted by Procrustes distance (p_d) from (a), while off-diagonal (i, j) entries give p_d between the i^{th} and j^{th} mappings. The matrices and mean $p_d \pm \text{std}$ values indicate that default UMAP embeddings are not self-similar across random starts, whereas the modified and composite attraction shapes promote scale-invariant structure.

A direct way to induce “far-sightedness” is to increase the attractive coefficient at large embedding distances, thereby allowing distant high-dimensional neighbors to move toward one another. To test this mechanism, we modify the UMAP attraction shape as follows:

$$f_a^M = f_a^U - \beta\zeta, \quad (13)$$

where β controls the magnitude of the added term (we used $\beta = 0.2$, Fig. 2(e)). Adding this term to the attraction shape is equivalent to adding a regularizer to the attractive component of the loss (i.e., $\mathcal{L}^M = \mathcal{L} + \sum_{i,j} \beta/3 \|y_i - y_j\|_2^2$). Beyond increasing attraction for distant pairs, the modification promotes point intermixing during early optimization, which improves convergence from random initialization in a manner analogous to early exaggeration in t -SNE. Equation (13) uses the simplest linear correction; alternative distance functions, such as $\log \zeta$ or ζ^p ($p \in \mathbb{R}_{\geq 0}$), may provide similar control.

We also evaluate a composite attraction shape:

$$f_a^C = \begin{cases} f_a^M, & \text{epoch} \leq 100 \\ f_a^U, & \text{otherwise} \end{cases}. \quad (14)$$

The composite shape uses the modified attraction only during the initial epochs and then reverts to the original UMAP shape, with the goal of reducing distortions introduced by f_a^M . We next evaluate how the modified and composite shapes affect dimensionality reduction from random initialization.

We first construct a PCA-initialized MNIST embedding as the reference mapping in Figure 2(a). For each attraction shape, we then generate Gaussian-randomly initialized embeddings and repeat the procedure 100 times. We quantify alignment using Procrustes analysis (Gower, 1975), which registers two point clouds under scaling, translation, rotation, and reflection (for details, see Appendix D.1). Specifically, each randomly initialized embedding is aligned to the PCA-initialized reference, and the residual discrepancy is measured by the Procrustes distance (p_d). Figure 2(b) shows the four UMAP embeddings with the smallest p_d . Their cluster shapes are similar, but their cluster placements differ. The modified and composite attraction shapes produce more consistent cluster placements (Figs. 2(c) and (d), respectively).

To characterize placement consistency more completely, we compute a Procrustes matrix. Its diagonal entries are sorted by p_d relative to the PCA-initialized mapping, and its off-diagonal entries contain p_d values between pairs of randomly

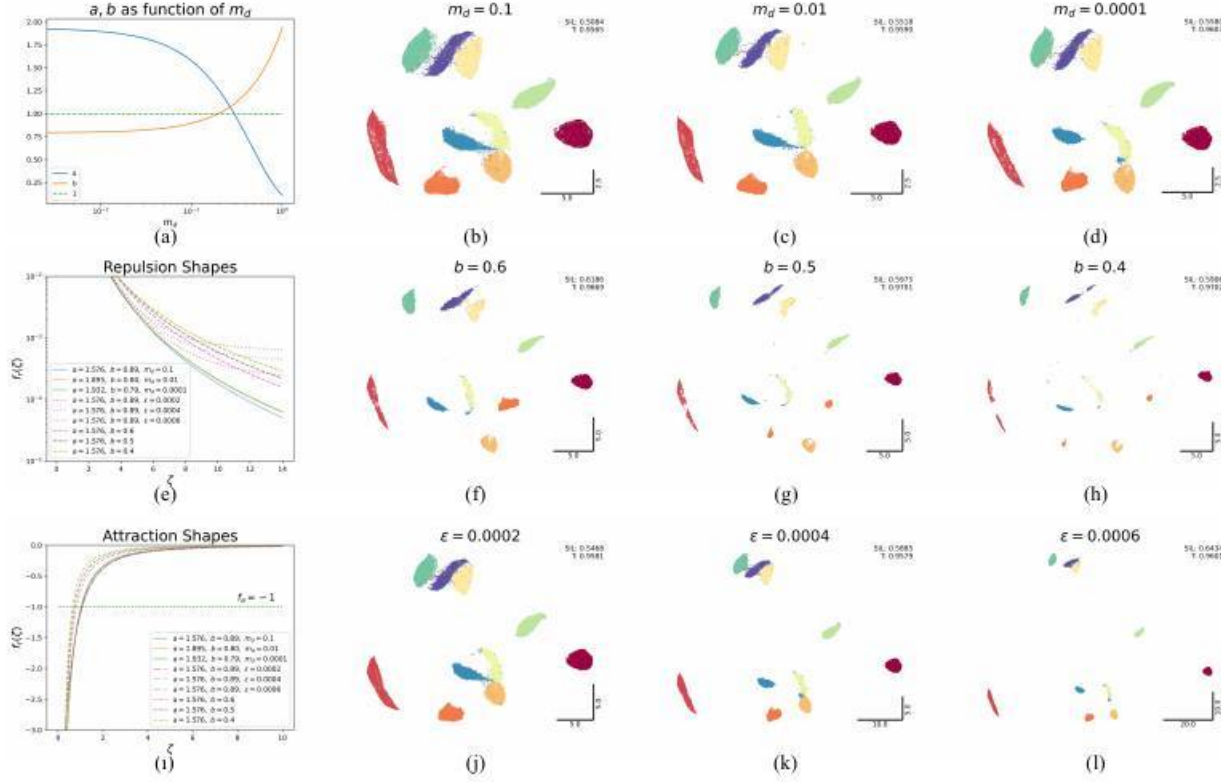


Figure 3: Inter-cluster-distance control on the MNIST dataset. (a) Estimating a, b by varying the low-dimensional distance m_d restricts the accessible parameter region. (b-d) UMAP outputs for $m_d = 0.1, 0.01$, and 0.0001 show limited improvement in cluster compactness. (e) Repulsion shapes induced by different parameter settings. (f-h) Explicitly increasing repulsion by varying b yields more compact clusters and reveals clusters that are otherwise absent. (i) Attraction shapes under parameter variation. (j-l) Adding a small positive value (ϵ) to the repulsion shape increases inter-cluster distance.

initialized mappings (for details, see Appendix D.1). This construction is analogous to a similarity matrix (Foote, 1999). Embeddings produced with the default UMAP attraction shape are mutually dissimilar (Fig. 2(f)), whereas embeddings produced with the modified (Fig. 2(g)) and composite (Fig. 2(h)) shapes exhibit strong mutual similarity. These results indicate that UMAP encodes a reproducible structure that can be recovered independently of initialization, but that recovery can fail when long-distance attraction is too weak.

5.2 Cluster Formation and Compactness

In UMAP, the principal user-facing parameter associated with cluster formation is the minimum-distance parameter m_d through Eq. 2. However, changing m_d explores only a restricted subset of the corresponding a, b parameter space (Fig. 3(a)). Consequently, decreasing m_d often yields embeddings with little additional compactness benefit (Figs. 3(b-d)). The limiting factor is that m_d only weakly changes the repulsion shape (Fig. 3(e)). A more direct intervention is to vary a and b explicitly. Decreasing a increases repulsion, but it decreases attraction even more rapidly and therefore intensifies near-sightedness. Decreasing b provides more effective control (Figs. 3(e,i)). Figures 3(f-h) show that setting b to $0.6, 0.5$, and 0.4 progressively increases inter-cluster separation and splits previously merged clusters. This splitting is consistent with the increasingly heavy-tailed kernel obtained as b decreases; heavy-tailed kernels are known to produce smaller, more distinct clusters (Van der Maaten and Hinton, 2008; Kobak et al., 2019), and Kobak et al. (2019) discusses how b modulates the UMAP tail. Although this intervention separates all ten MNIST labels into distinct clusters, it does not isolate attraction from repulsion: changing b amplifies repulsion more than changing m_d , but it also changes the attraction profile.

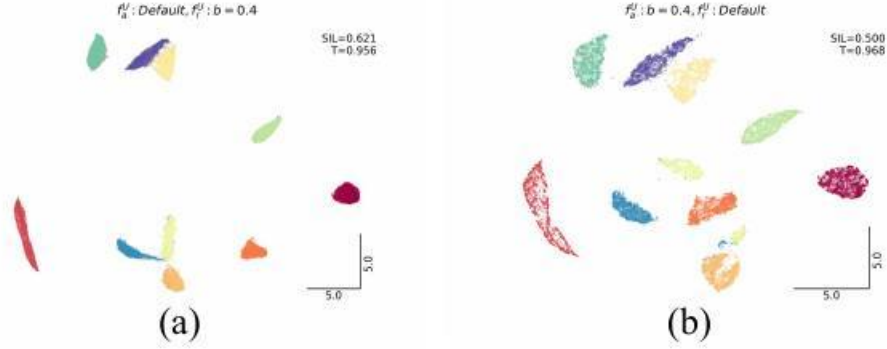


Figure 4: MNIST embeddings under cross-composed shapes. (a) Default attraction shape paired with the repulsion shape obtained at $b = 0.4$. (b) Default repulsion shape paired with the attraction shape obtained at $b = 0.4$. The first configuration preserves the default UMAP cluster identities while increasing compactness, whereas the second introduces additional intra-cluster structure and new clusters.

To isolate the effect of repulsion, we hold the attraction shape fixed and modify only the repulsion shape by adding a small positive constant (ε):

$$f_r^M = f_r^U + \varepsilon, \quad (15)$$

with a and b held fixed. This modification is equivalent to adding a regularizer to the repulsive term of the loss (i.e., $\mathcal{L}^M = \mathcal{L} - \sum_{i,j} \varepsilon/2 \|y_i - y_j\|_2^2$), and it yields stronger repulsion than the standard parameterization permits (Fig. 3(e)). Figs. 3(j-l) show that increasing ε increases inter-cluster distance. The resulting clustering behavior nevertheless resembles the behavior obtained by varying m_d , with loose separation of all labels as ε increases. Because ε leaves the attraction shape unchanged, whereas varying m_d traces similar attraction shapes (Fig. 3(i)), these results suggest that cluster formation is governed primarily by attraction.

To separate the mechanisms underlying cluster formation, we vary either the attraction shape or the repulsion shape while leaving the other at its default, using the shape induced by setting b to 0.4 (Fig. 4). When the default attraction shape is paired with the stronger repulsion shape, no new structures or clusters appear, although the existing UMAP clusters become smaller (Fig. 4(a)). Conversely, when the stronger attraction shape is paired with the default repulsion shape, the embedding develops additional structure within existing clusters (Fig. 4(b)). Several original clusters also split into smaller clusters. These cross-compositions indicate that attraction is the primary driver of cluster formation, whereas repulsion controls compactness either by shrinking clusters or by increasing inter-cluster distance, depending on the repulsion shape. Appendix G.3 shows that LocalMAP embeddings exploit the same attraction–repulsion interplay to separate clusters.

6 Comparison to NEG- t -SNE

Many dimensionality-reduction algorithms use the parameter setting $a = 1$ and $b = 1$. This choice often yields stable, bounded gradients and therefore simplifies optimization. Recently, Damrich et al. (2023) analyzed this setting for UMAP and introduced Neg- t -SNE, which combines stable gradients with compact clustering even when the learning rate is fixed at 1. Parametric UMAP (Sainburg et al. 2021) initially used the original UMAP loss, but later adopted a numerically stable modified cross-entropy objective (Shi et al. 2023) with a logsigmoid kernel.¹ Analyzing both objectives shows that they arrive at the same formulation from different starting points, giving the following observation:

Proposition 6.1. *Neg- t -SNE is Parametric UMAP with $a = 1$ and $b = 1$.*

For $a = 1$ and $b = 1$, the UMAP attraction and repulsion shapes are $f_a^U = -2/(1 + \zeta^2)$ and $f_r^U = 2/(\zeta^2(1 + \zeta^2))$, respectively. The attraction shape is bounded in $[-2, 0]$ and satisfies $f_a^U(0) = -2$ (Fig. 1(f)), whereas the repulsion shape remains effectively unchanged and is still unbounded as $\zeta \rightarrow 0$ (Fig. 1(g)). Because $f_a^U < -1$ near $\zeta = 0$, Proposition 4.1 and Section 5 imply that this unity-parameter case still requires learning-rate annealing.

Damrich et al. (2023) analyzed UMAP’s negative-sampling loss through the contrastive-embedding (CE) framework and identified $1/\zeta^2$ as the effective UMAP kernel. Within the CE formulation, NEG- t -SNE is obtained by replacing

¹github.com, merged on April 26, 2022

this kernel with $1/(1 + \zeta^2)$. Expressed in UMAP notation, this substitution gives the low-dimensional affinity $q_{ij}^N = 1/(2 + \zeta^2)$. The corresponding attraction and repulsion shapes are $f_a^N = -2/(2 + \zeta^2)$, and $f_r^N = 2/((1 + \zeta^2)(2 + \zeta^2))$, respectively (Figs. 1(f,g)). The attraction shape is bounded in $[-1, 0]$ and therefore satisfies Proposition 4.1. Any $\lambda \in [0, 1]$ produces contraction without inducing contraction–expansion oscillations. Consequently, NEG- t -SNE is less sensitive to learning-rate annealing, and its clusters remain less diffuse even at constant $\lambda = 1.0$. Its repulsion shape is also bounded in $[0, 1]$ and does not diverge as $\zeta \rightarrow 0$, so randomly sampled repulsive updates are less likely to eject points from clusters. Although Damrich et al. (2023) attributed this difference primarily to bounded NEG- t -SNE repulsion versus unbounded UMAP repulsion, the analysis in Fig. 1 shows that attraction is also a central determinant of cluster stability and compactness. Appendix F provides additional illustration. Together with Fig. 4, these results suggest that attraction and repulsion shapes can be combined across methods; for example, one can pair the UMAP attraction shape with the NEG- t -SNE repulsion shape. Appendix J reports examples using UMAP, NEG- t -SNE, and PaCMAP.

7 Discussion and Conclusion

This work analyzed how attractive and repulsive update terms determine cluster formation in neighbor-embedding algorithms. Although these terms are commonly described as pulling similar points together and pushing dissimilar points apart, their distance-dependent mechanisms have not been fully specified. The shape-based analysis developed here formalizes several aspects of the cluster-formation dynamics. The main text focuses on UMAP and NEG- t -SNE, but the same framework applies more broadly to dimensionality reduction; additional algorithms are discussed in Appendix G.

We modeled attraction and repulsion coefficients as functional mappings of embedding distance and analyzed their shapes. The resulting characterization (Propositions 4.1 and 4.2) reveals a counterintuitive regime: at sufficiently short distances, the UMAP attractive update produces distance expansion rather than contraction (Fig. 1(b)). UMAP’s learning-rate schedule mitigates this regime by driving the effective contraction threshold toward zero.

We also examined the initialization dependence of the low-dimensional embedding. When similar points are initialized far apart, the default attraction shape can produce insufficient contraction. The proposed formalism suggests an explicit remedy: add terms to the attraction shape that increase long-distance attraction. This modification improves output consistency under random initialization, supporting the interpretation that UMAP encodes a reproducible structure and that initialization and scheduling mechanisms primarily help the optimizer reach that structure.

The repulsion-shape analysis shows that stronger repulsion increases inter-cluster distance. We therefore added a small positive constant to the repulsion shape, producing a level of repulsion that is not attainable by varying the standard minimum-distance parameter m_d alone.

Finally, we characterized the relationship between NEG- t -SNE and UMAP. In the special case $a = 1$ and $b = 1$, default NEG- t -SNE differs primarily by using bounded attraction ($\in [-1, 0]$) and bounded repulsion ($\in [0, 1]$) shapes. Both propositions are then satisfied across the fixed-learning-rate range $\in [0, 1]$, which explains the method’s reduced sensitivity to learning-rate choice. Allowing a and b to vary in NEG- t -SNE reintroduces the same numerical issues observed in UMAP. We also showed that NEG- t -SNE and Parametric UMAP are equivalent under the stated formulation.

These results provide technical tools for designing and optimizing dimensionality-reduction algorithms through explicit control of attraction and repulsion shapes. Because dimensionality reduction is closely related to contrastive learning (Damrich and Hamprecht, 2021; Hu et al., 2023), the same shape-based perspective may also inform representation-learning objectives. Overall, the analysis supports more principled, consistent, and reliable interpretation of neighbor embeddings.

Software and Data

All datasets used in this study are publicly available. The MNIST dataset is available at yann.lecun.com. Fashion-MNIST is available at github.com. Single-cell transcriptomes data is available at github.com.

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Appendix

Appendix A gives the derivations for the UMAP attraction and repulsion shapes. Appendix B extends the main-text discussion of constant learning rates. Appendix C evaluates the implications of strengthened attraction and repulsion for large-language-model embeddings. Appendix D.1 defines the Procrustes distance and Procrustes matrix used for initialization-consistency analysis. Appendix E repeats the random-initialization experiment on additional datasets. Appendix F further analyzes the relationship between NEG- t -SNE and UMAP, and Appendix G applies the shape framework to alternative dimensionality-reduction algorithms. Appendix H specifies the construction of the high-dimensional graphs. Appendix I reports the full sweeps over λ_a and λ_r from Fig. 1 for UMAP, NEG- t -SNE, and PaCMAP on multiple datasets. Appendix J examines cross-method combinations of attraction and repulsion shapes. Appendix K lists the implementation details.

A Proofs and Derivations

A.1 Proof of Proposition 4.1

Proof. Subtracting Eq. (7) from Eq. (8) and taking the norm gives

$$\|y_i^{t+1} - y_j^{t+1}\| = |1 + 2\lambda f_a| \|y_i^t - y_j^t\|. \quad (16)$$

The distance contracts precisely when $|1 + 2\lambda f_a| < 1$, equivalently when

$$-1 < \lambda f_a < 0, \quad (17)$$

□

A.2 Proof of Proposition 4.2

Proof. Subtracting Eq. (9) from Eq. (10) and taking the norm gives

$$\|y_i^{t+1} - y_j^{t+1}\| = |1 + \lambda f_r| \|y_i^t - y_j^t\|. \quad (18)$$

The distance expands when $|1 + \lambda f_r| > 1$, equivalently

$$\lambda f_r < -2 \text{ or } f_r > 0. \quad (19)$$

From the definition of f_r , the latter suffices. □

A.3 Attraction Shape:

To derive the attraction shape, we use the general low-dimensional affinity $q_{ij} = (\gamma + a\|y_i - y_j\|_2^{2b})^{-1}$. This form reduces to UMAP when $\gamma = 1$ and to NEG- t -SNE when $\gamma = 2$. The attractive force is

$$\begin{aligned} \nabla_{y_i} \log q_{ij} &= -\nabla_{y_i} \log (\gamma + a(\|y_i - y_j\|_2^2)^b) \\ &= -\frac{1}{\gamma + a(\|y_i - y_j\|_2^2)^b} \nabla_{y_i} (\gamma + a(\|y_i - y_j\|_2^2)^b) \\ &= -\frac{1}{\gamma + a(\|y_i - y_j\|_2^2)^b} ab(\|y_i - y_j\|_2^2)^{b-1} \nabla_{y_i} \|y_i - y_j\|_2^2 \\ &= -\frac{2ab(\|y_i - y_j\|_2^2)^{b-1}}{\gamma + a(\|y_i - y_j\|_2^2)^b} (y_i - y_j). \end{aligned} \quad (20)$$

After defining $\zeta = \|y_i - y_j\|_2$, the scalar coefficient defines the attraction shape:

$$f_a(\zeta) = -\frac{2ab\zeta^{2(b-1)}}{\gamma + a\zeta^{2b}} \quad (21)$$

A.4 Condition for strictly increasing f_a^U :

The distance dependence of f_a^U is characterized by its derivative:

$$\frac{df_a^U(\zeta)}{d\zeta} = -\frac{2ab\zeta^{2b-3}}{\gamma + a\zeta^{2b}} \left((b-1) - \frac{ab\zeta^{2b}}{\gamma + a\zeta^{2b}} \right). \quad (22)$$

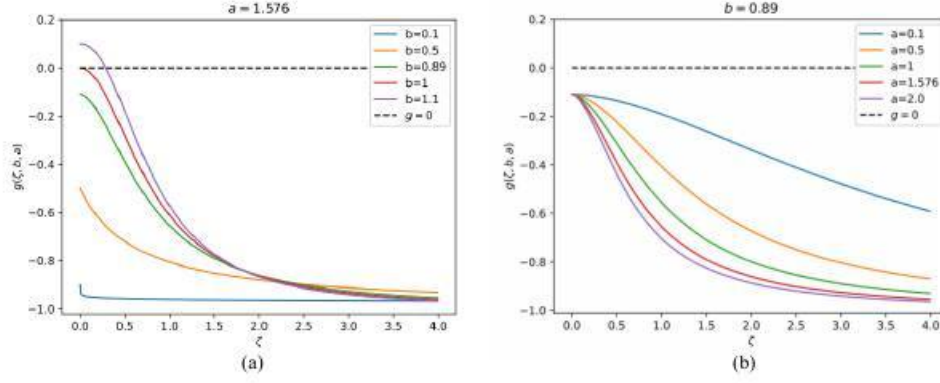


Figure 5: Evaluation of $g(\zeta, b, a)$ under two one-parameter sweeps: (a) a fixed at 1.576 and (b) b fixed at 0.89.

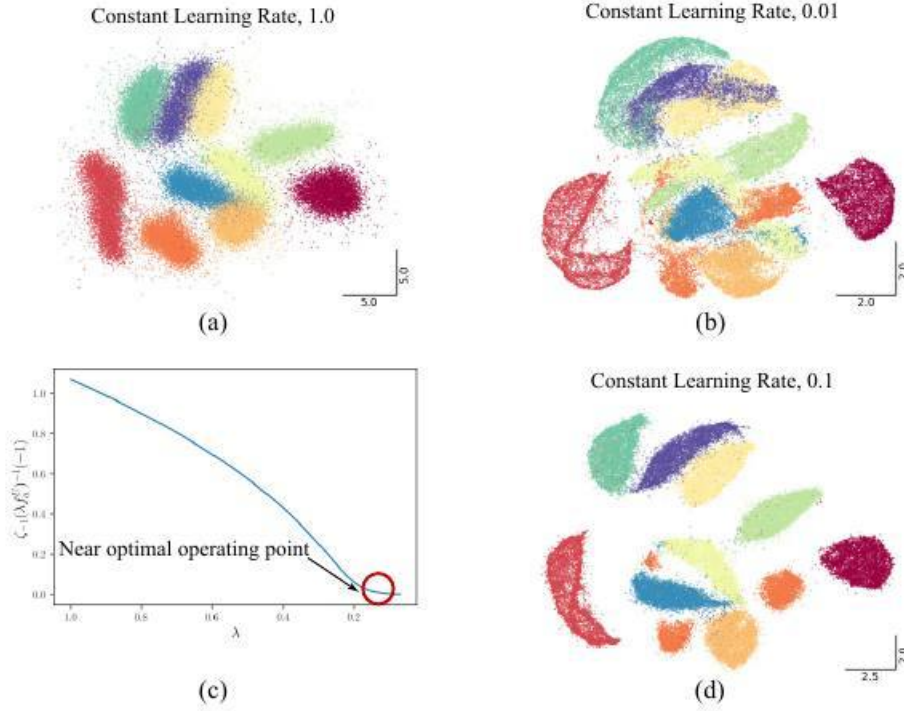


Figure 6: Effect of a constant learning rate on embeddings. (a) At a high learning rate ($\lambda = 1.0$), embeddings are diffuse because ζ_{-1} is large. (b) At a low learning rate ($\lambda = 0.01$), clusters do not form because both attraction and repulsion are too weak. (c) ζ_{-1} decreases nonlinearly as the learning rate decreases. Effective optimization reduces ζ_{-1} while preserving sufficient attraction and repulsion magnitudes. (d) Distinct, compact clusters form at the near-optimal constant learning rate $\lambda = 0.1$.

This derivative yields the strictly increasing condition ($\frac{df^U}{d\zeta} > 0$),

$$g(\zeta, b, a) < 0, \quad (23)$$

where $g(\zeta, b, a) = b - 1 - \frac{ab\zeta^{2b}}{\gamma + a\zeta^{2b}}$. The inequality holds for $0 < b \leq 1$, as follows from the derivative and asymptotic behavior of g . Figure 5 evaluates g over representative values of b and a . When $b > 1$, condition (23) no longer holds.

A.5 Repulsion Shape:

Using the general low-dimensional affinity, the repulsive force is

$$\begin{aligned}
 \nabla_{y_i} \log(1 - q_{ij}) &= \nabla_{y_i} \log \left[1 - \frac{1}{\gamma + a(\|y_i - y_j\|_2^2)^b} \right] \\
 &= \frac{\gamma + a(\|y_i - y_j\|_2^2)^b}{(\gamma - 1) + a(\|y_i - y_j\|_2^2)^b} \nabla_{y_i} \left[1 - \frac{1}{\gamma + a(\|y_i - y_j\|_2^2)^b} \right] \\
 &= \frac{1}{\gamma - 1 + a(\|y_i - y_j\|_2^2)^b} \frac{ab(\|y_i - y_j\|_2^2)^{b-1}}{\gamma + a(\|y_i - y_j\|_2^2)^b} \nabla_{y_i} \|y_i - y_j\|_2^2 \\
 &= \frac{2ab(\|y_i - y_j\|_2^2)^{b-1}}{(\gamma - 1 + a(\|y_i - y_j\|_2^2)^b)(\gamma + a(\|y_i - y_j\|_2^2)^b)} (y_i - y_j).
 \end{aligned} \tag{24}$$

The scalar coefficient defines the repulsion shape:

$$f_r(\zeta = \|y_i - y_j\|_2) = \frac{2ab\zeta^{2(b-1)}}{(\gamma - 1 + a\zeta^{2b})(\gamma + a\zeta^{2b})}. \tag{25}$$

Thus, in the relevant domain, $f_r > 0$.

A.6 Loss functions due to modified attraction and repulsion shape:

The modified attraction term in Eq. (13) corresponds to the cost contribution

$$- \int (f_a^U(\|y_i - y_j\|_2) - \beta\|y_i - y_j\|_2)(y_i - y_j) dy_i = -\log(q_{ij}) + \frac{\beta}{3}\|y_i - y_j\|_2^3, \tag{26}$$

whereas the repulsion modification in Eq. (15) corresponds to

$$- \int (f_r^U(\|y_i - y_j\|_2) + \epsilon)(y_i - y_j) dy_i = -\log(1 - q_{ij}) - \frac{\epsilon}{2}\|y_i - y_j\|_2^2. \tag{27}$$

In both cases, the added component functions as a norm-based regularizer. Expressing the modification directly at the level of attraction and repulsion shapes makes its role explicit: the attraction term increases far-sightedness, whereas the repulsion term adds a constant repulsive coefficient.

B More on UMAP’s Learning Rate

Sections 5 and 6 show that UMAP requires learning-rate annealing under its default shape configuration (Fig 6). Section 5 introduced the minimum contraction distance ζ_{-1} and showed that reducing this value through learning-rate annealing yields compact clusters. Section 6 then compared the attraction shapes of UMAP with $a = 1.0$ and $b = 1.0$ against Neg- t -SNE, showing that Neg- t -SNE tolerates a constant learning rate of 1.0 better because its attraction shape lies in $[-1, 0]$, whereas UMAP’s lies in $[-2, 0]$. By the same argument, UMAP tolerates a constant learning rate of 0.5 when the effective attraction shape is confined to $[-1, 0]$ (Figs. 11(f,h)).

For a broad range of a and b values, however, annealed learning rates still produce better embeddings. The reason is that the optimization should eventually drive ζ_{-1} to zero, which occurs only as the learning rate approaches zero. If this reduction does not occur, the embedding remains diffuse (Fig 6(a)). Conversely, if the learning rate is too small from the beginning, attraction and repulsion are both too weak to produce clear clusters (Fig 6(b)). The ζ_{-1} - λ curve (Fig 6(c)) identifies $\lambda = 0.1$ as a near-optimal constant value because ζ_{-1} is low while attraction and repulsion retain sufficient magnitude. At this constant learning rate, the embedding forms compact clusters with clear boundaries (Fig 6(d)).

C Large Language Model Embedding

Strengthening attraction and repulsion can improve embedding quality in large text-representation settings. To evaluate this effect, we randomly selected 300,000 PubMedBERT (Gu et al., 2021) feature vectors with known labels from the PubMed dataset (González-Márquez et al., 2024). Default UMAP yields a crowded embedding in which label colors overlap substantially (Fig. 7(a)). Without labels, the map provides limited visual evidence of distinct clusters. Setting $b = 0.6$, as motivated in Section 5.2, increases repulsion while simultaneously improving attraction-mediated cluster formation. The resulting embedding remains interpretable even without explicit labels (Fig. 7(b)). We quantify this improvement using k-NN accuracy with $k = 10$. On a hold-out set of 10,000 points, reducing b to 0.6 increases k-NN accuracy from 49.2% to 55.48%.

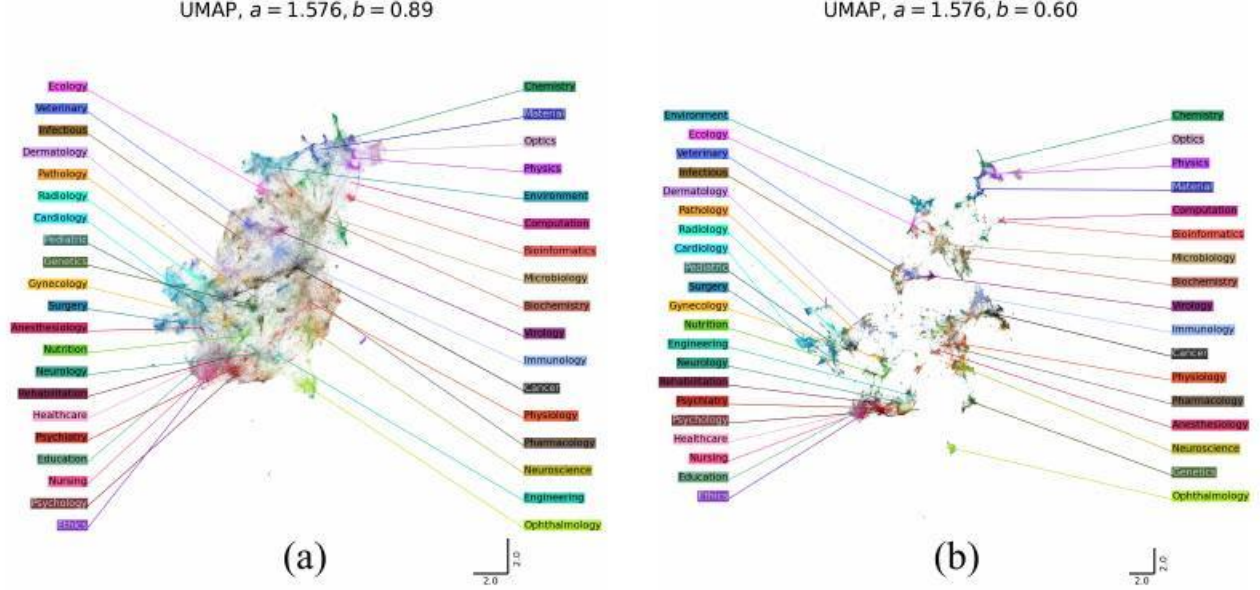


Figure 7: Visualization of 300K PubMed abstracts. (a) Default UMAP ($a = 1.576, b = 0.89$) produces substantial label overlap in the embedding. (b) UMAP with strengthened repulsion and attraction ($a = 1.576, b = 0.60$) produces more separated clusters and non-overlapping labels. Labels are displayed in both panels to support interpretation.

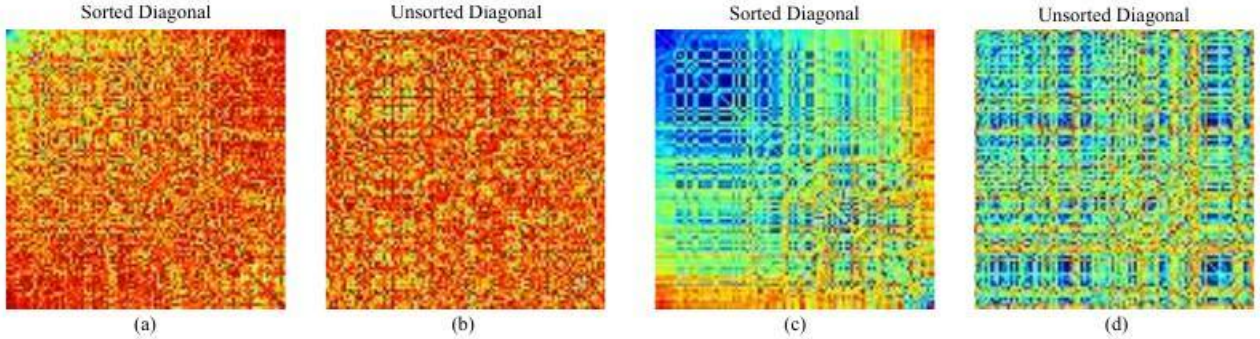


Figure 8: Visualization effect of sorting the Procrustes-matrix diagonal. (a) Procrustes matrix from [2]f for the default UMAP attraction shape, with diagonal entries sorted by p_d relative to a PCA-initialized reference embedding. (b) Same matrix entries as in (a), but with the diagonal unsorted (or randomly sorted). (c) Procrustes matrix from [2]g for the modified attraction shape, with the diagonal sorted by reference p_d . (d) Same entries as in (c), but with the diagonal unsorted.

D Metrics

D.1 Procrustes Distance and Procrustes Matrix

The Procrustes distance [Gower, 1975] measures the discrepancy between two point clouds $\{x\}$ and $\{y\}$ after admissible linear transformations, namely translation, scaling, and rotation. Operationally, $\{x\}$ is fixed and $\{y\}$ is transformed to maximize alignment. Let $\{y'\}$ denote the transformed version of $\{y\}$ that achieves this alignment. The Procrustes distance is then

$$p_d(\{x\}, \{y\}) = \sqrt{\sum_k (x_k - y'_k)^2}. \quad (28)$$

This linear discrepancy measure has been useful in several embedding-analysis settings [McInnes et al., 2018; Islam and Fleischer, 2022; Kotlov et al., 2024].

We use Procrustes distance to quantify embedding consistency under random initialization. Let $\{x\}_p$ denote the reference embedding, which is PCA-initialized in our experiments, and let $X_r = \{\{x\}_i | i = 1, 2, 3, \dots, N\}$ denote N embeddings obtained from random initialization. Embedding similarity is summarized by the mean and standard deviation of the strictly lower-triangular entries of the Procrustes matrix P (reported as mean $\pm$ std in Figs. 2, 9 and 10), with

$$\text{mean} = \frac{2}{N(N-1)} \sum_{i,j(i>j)} P_{i,j}, \text{ and } \text{std} = \sqrt{\sum_{i,j(i>j)} \frac{2(P_{i,j} - \text{mean})^2}{N(N-1)}} \quad (29)$$

The embeddings in X_r can be ordered so that $p_d(\{x\}_i, \{x\}_p) \leq p_d(\{x\}_{i+1}, \{x\}_p)$. Under this ordering, the diagonal entries of the Procrustes matrix are

$$P_{i,i} = p_d(\{x\}_i, \{x\}_p) \quad (30)$$

and the off-diagonal values are given by

$$P_{i,j} = p_d(\{x\}_i, \{x\}_j). \quad (31)$$

Sorting the diagonal does not change the numerical values, but it makes the self-similarity structure visible. In Fig. 8(a), for example, the sorted diagonal concentrates mutually similar embeddings in the upper-left region of the matrix. Under the modified attraction shape, the set of mutually similar embeddings expands, as indicated by the larger blue region in Fig. 8(c); this pattern suggests a metastable embedding state. When the diagonal is unsorted, the same structure is visually obscured (Fig. 8(b,d)).

D.2 Trustworthiness

The trustworthiness metric (Venna and Kaski 2001) quantifies local-neighborhood preservation after dimensionality reduction:

$$T = 1 - \frac{2}{nk(2n - 3k - 1)} \sum_{i=1}^n \sum_{y_j \in \text{KNN}(y_i, k)} \max(0, r(i, j) - k) \quad (32)$$

where $\text{KNN}(y_i, k)$ denotes the k -NN graph in embedding space and $r(i, j)$ denotes the rank of x_j in the high-dimensional k -NN graph. In practice, k is often set to 5, so the metric evaluates preservation of each point's five nearest neighbors. For computational efficiency, all reported trustworthiness values use 10,000 randomly sampled indices, with the same index set used across embeddings in each comparison.

D.3 silhouette score

Although the silhouette score (Rousseeuw 1987) was designed for clustering evaluation, we use it here to quantify label separation in the embedding, i.e., how well ground-truth labels are separated by the learned coordinates. The metric is appropriate because these embedding algorithms produce visually clustered structures whose alignment with labels is of interest. For a point y_i in the point cloud $\{y\}$, let a_i be the mean distance from y_i to points with the same label, and let b_i be the smallest mean distance from y_i to points with any other label. The pointwise silhouette is

$$s_i = \frac{b_i - a_i}{\max\{a_i, b_i\}} \quad (33)$$

This value lies in $[-1, 1]$. Values close to 1 indicate that y_i is well placed within its own label cluster, values near 0 indicate boundary points, and values close to -1 indicate poor label separation. The overall silhouette score is

$$\text{SIL} = \frac{1}{N} \sum s_i. \quad (34)$$

We compute the silhouette score over the full embedding, without random sampling, using Euclidean distances.

E Effect of Random Initialization in Additional Datasets

Section 5.1 reports the random-initialization experiment on MNIST. Here we repeat the same protocol on Fashion-MNIST (FMNIST) (Xiao et al. 2017) and single-cell transcriptomes (Macosko et al. 2015) (Fig. 9 and 10 respectively). The conclusion is unchanged: modified and composite attraction shapes that increase long-distance attraction substantially improve reconstruction consistency.

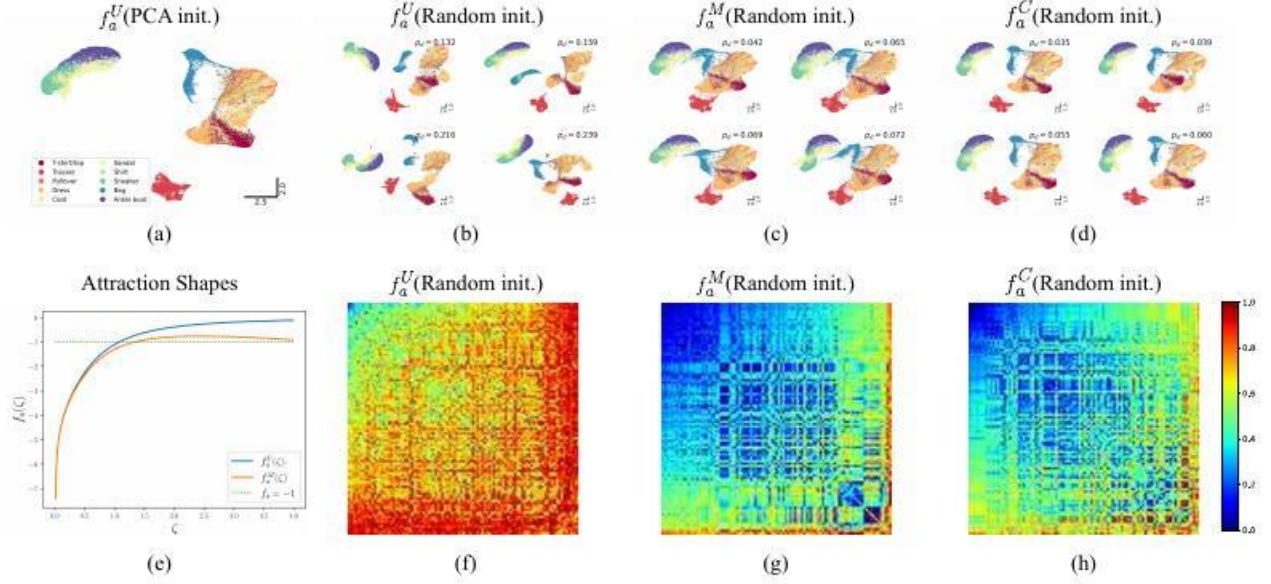


Figure 9: Effect of random UMAP initialization under different attraction shapes on FMNIST. (a) PCA-initialized reference mapping. (b-d) Four mappings with the smallest Procrustes distance (p_d) from (a), shown for (b) default, (c) modified, and (d) composite attraction shapes. (e) Default UMAP and modified attraction shapes. (f-h) Procrustes matrices from 100 runs for (f) default (0.72 ± 0.15), (g) modified (0.38 ± 0.19), and (h) composite (0.43 ± 0.18) attraction shapes.

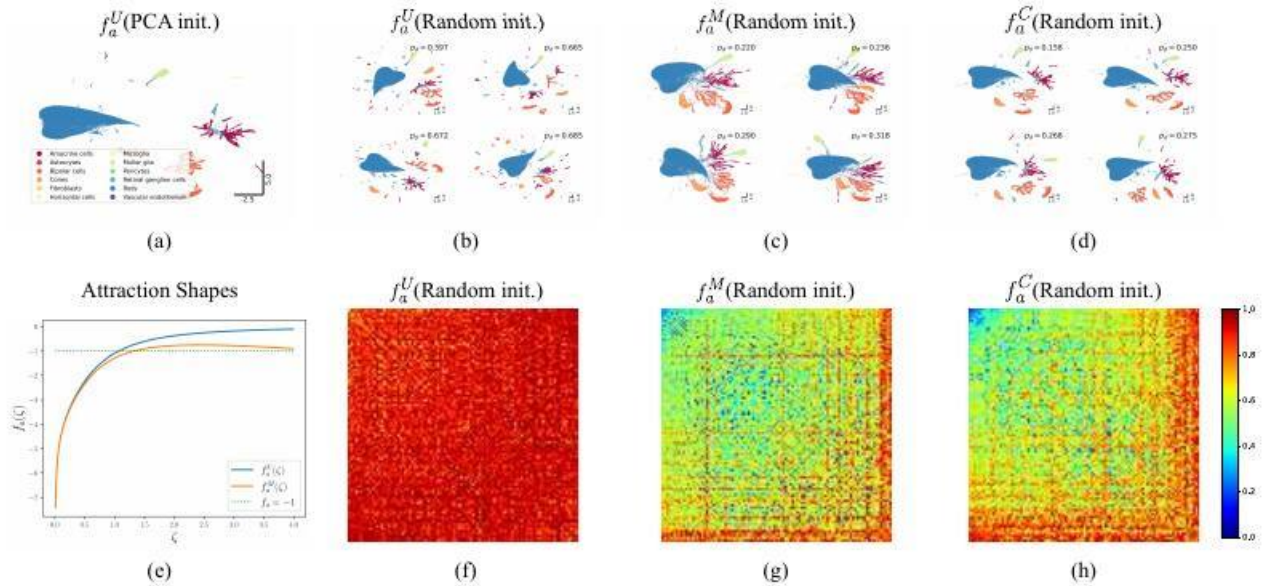


Figure 10: Effect of random UMAP initialization under different attraction shapes on single-cell transcriptomes. (a) PCA-initialized reference mapping. (b-d) Four mappings with the smallest Procrustes distance (p_d) from (a), shown for (b) default, (c) modified, and (d) composite attraction shapes. (e) Default UMAP and modified attraction shapes. (f-h) Procrustes matrices from 100 runs for (f) default (0.91 ± 0.06), (g) modified (0.61 ± 0.13), and (h) composite (0.64 ± 0.15) attraction shapes.

F More on NEG- t -SNE

F.1 Comparison to UMAP

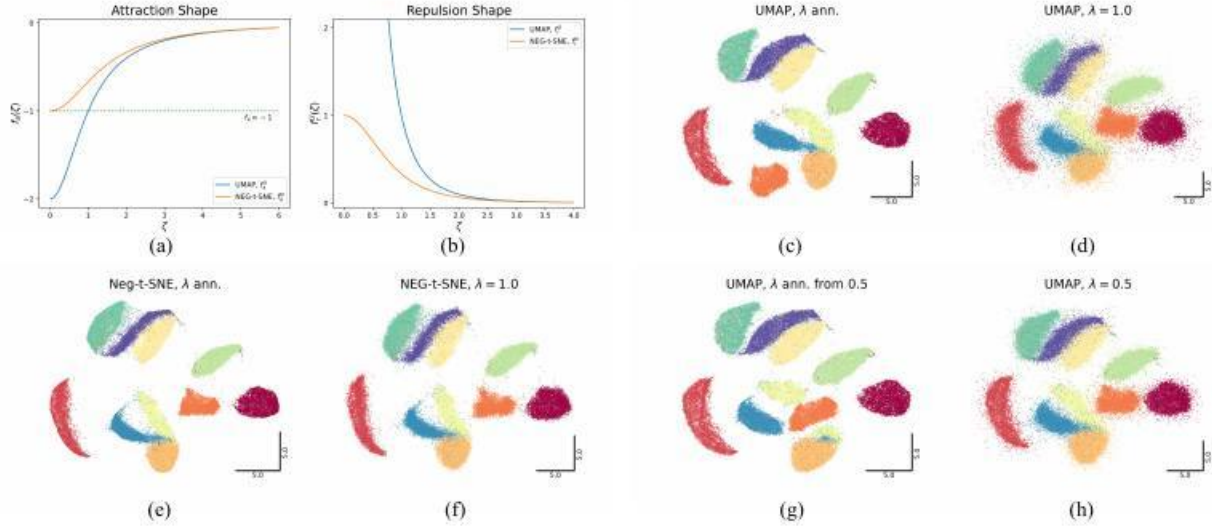


Figure 11: Learning-rate sensitivity of UMAP and NEG- t -SNE on MNIST. (a) Attraction and (b) repulsion shapes for UMAP ($a = 1$, $b = 1$) and NEG- t -SNE. (c,d) UMAP is sensitive to λ because $f_a^U < -1$ at small separation distances ζ ; without annealing, clusters become diffuse. (e,f) NEG- t -SNE is less sensitive to λ because $f_a^N \in [-1, 0]$ for all distances, yielding less diffuse clusters without annealing. (g,h) Setting $\lambda = 0.5$ confines the effective f_a^U to $[-1, 0]$ and reduces learning-rate sensitivity.

Figure 11 compares the UMAP and NEG- t -SNE shapes together with MNIST embeddings. With annealed learning rates, the two methods produce similar embeddings (Figs. 11(c,e)). With a constant learning rate of 1, UMAP produces a diffuse structure, whereas NEG- t -SNE yields cleaner cluster boundaries (Fig. 11(d,f)). Section 6 implies that constraining f_a^U to $[-1, 0]$ should reduce cluster diffuseness at fixed λ . This pattern also appears in Fig. 11(h,i), where UMAP gives better embedding and clustering when the attraction learning rate (λ_a) is $\lesssim 0.5$. A direct implementation is to initialize λ at 0.5, which satisfies Proposition 4.1 for all ζ . The embeddings in Figs. 11(g) and (h) confirm that the resulting clusters resemble those from NEG- t -SNE; at constant $\lambda = 0.5$, they are less diffuse and have sharper boundaries than the $\lambda = 1$ UMAP embeddings (Fig. 11(d,h)). A small number of points still remain outside clusters because of UMAP’s repulsion shape, a behavior mitigated by NEG- t -SNE.

The parameters a and b can also be introduced into NEG- t -SNE, yielding the Parametric UMAP formulation described in Proposition 6.1. The affinity becomes $q_{ij}^N = 1/(2 + a\zeta^{2b})$, and the attraction and repulsion shapes become $f_a^N = -\frac{2ab\zeta^{2(b-1)}}{2+a\zeta^{2b}}$, and $f_r^N = \frac{2ab\zeta^{2(b-1)}}{(1+a\zeta^{2b})(2+a\zeta^{2b})}$, respectively. For $0 < b < 1$, both shapes diverge as $\zeta \rightarrow 0$. Therefore, allowing a and b to vary gives NEG- t -SNE numerical challenges analogous to those of UMAP. A notable distinction is that the attraction shape reaches a smaller minimum contraction distance (ζ_{-1}) than UMAP. This can improve cluster formation, but the attraction also approaches zero more rapidly at large distances, increasing near-sightedness and potentially weakening long-range attraction.

F.2 Comparison to Parametric UMAP

Parametric UMAP was originally trained with the standard UMAP objective (Sainburg et al. 2021). Later work adopted a numerically stable modified cross-entropy loss based on a log-sigmoid kernel (Shi et al. 2023). This modification makes Parametric UMAP equivalent to Neg- t -SNE (Damrich et al. 2023) under Proposition 6.1; the equivalence is shown below.

Proof of Proposition 6.1

Proof. The kernel function under this modification becomes

$$q_{ij}^P = -\log(1 + a\|y_i - y_j\|_2^{2b}). \quad (35)$$

The attractive term is

$$-\text{logsigmoid}(q_{ij}^P) = -\log\left(\frac{1}{1 + \exp(-q_{ij}^P)}\right) \quad (36)$$

$$= -\log\left(\frac{1}{2 + a\|y_i - y_j\|_2^{2b}}\right) \quad (37)$$

$$= -\log(q_{ij}^N). \quad (38)$$

And the repulsive term is

$$\text{logsidmoid}(q_{ij}^P) - q_{ij}^P = \log\left(\frac{1}{2 + a\|y_i - y_j\|_2^{2b}}\right) + \log(1 + a\|y_i - y_j\|_2^{2b}) \quad (39)$$

$$= \log\left(\frac{1 + a\|y_i - y_j\|_2^{2b}}{2 + a\|y_i - y_j\|_2^{2b}}\right) \quad (40)$$

$$= \log\left(1 - \frac{1}{2 + a\|y_i - y_j\|_2^{2b}}\right) \quad (41)$$

$$= \log(1 - q_{ij}^N). \quad (42)$$

Both terms match NEG-t-SNE with explicit parameters a and b ; in the default Neg- t -SNE formulation, these parameters are fixed to 1. $\square$

G Alternate Dimensionality Reduction Algorithms

The alternative algorithms considered here use the same low-dimensional kernel as UMAP when $a = 1$ and $b = 1$:

$$q_{ij} = \frac{1}{1 + \|y_i - y_j\|_2^2}. \quad (43)$$

This section first analyzes TriMap (Amid and Warmuth, 2019). Although TriMap is defined over triplets rather than pairwise interactions, it provides a useful intermediate case for studying coupled attraction and repulsion mechanisms that are more complex than those in UMAP. We then consider Pairwise Controlled Manifold Approximation (PaCMAP) (Wang et al., 2021) and its extension, Pairwise Controlled Manifold Approximation with Local Adjusted Graph (LocalMAP) (Wang et al., 2025), which modify TriMap-style objectives for pairwise interactions. The section then treats t -SNE (Van der Maaten and Hinton, 2008), briefly discusses SNE (Hinton and Roweis, 2002) with an alternative kernel, and concludes with multidimensional scaling (Borg and Groenen, 2007).

G.1 TriMap

TriMap (Amid and Warmuth, 2019) optimizes low-dimensional using a triplet loss

$$\mathcal{L}^T = \sum_{(i,j,k)} w_{ijk} \frac{1}{1 + \frac{q_{ij}}{q_{ik}}}, \quad (44)$$

where w_{ijk} denotes the triplet weight, y_j belongs to the high-dimensional k -nearest-neighbor set of y_i , and y_k is a distant point. Minimizing this objective is intended to attract y_i toward y_j while repelling y_i from y_k . The update equations are

$$y_i^{t+1} = y_i^t + \lambda f_a^T(\zeta_1^t, \zeta_2^t)(y_i^t - y_j^t) + \lambda f_r^T(\zeta_1, \zeta_2)(y_i^t - y_k^t), \quad (45)$$

$$y_j^{t+1} = y_j^t - \lambda f_a^T(\zeta_1^t, \zeta_2^t)(y_i^t - y_j^t), \quad (46)$$

$$y_k^{t+1} = y_k^t - \lambda f_r^T(\zeta_1^t, \zeta_2^t)(y_i^t - y_k^t), \quad (47)$$

where $\zeta_1 = \|y_i - y_j\|_2$ is the nearest-neighbor distance, $\zeta_2 = \|y_i - y_k\|_2$ is the distance to the faraway point, and f_a^T and f_r^T are the TriMap attraction and repulsion shapes. Unlike UMAP, the attractive and repulsive components are not separable, and each shape depends on two distance variables. The attraction shape is

$$f_a^T(\zeta_1, \zeta_2) = -\frac{2(1 + \zeta_2^2)}{(2 + \zeta_1^2 + \zeta_2^2)^2}, \quad (48)$$

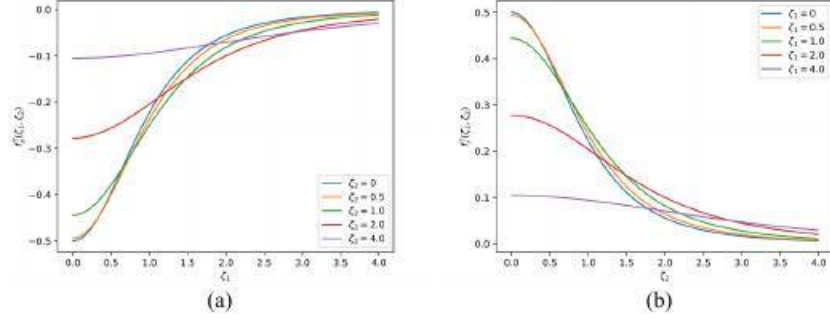


Figure 12: TriMap shape functions: (a) attraction shapes under varying ζ_2 and (b) repulsion shapes under varying ζ_1 .

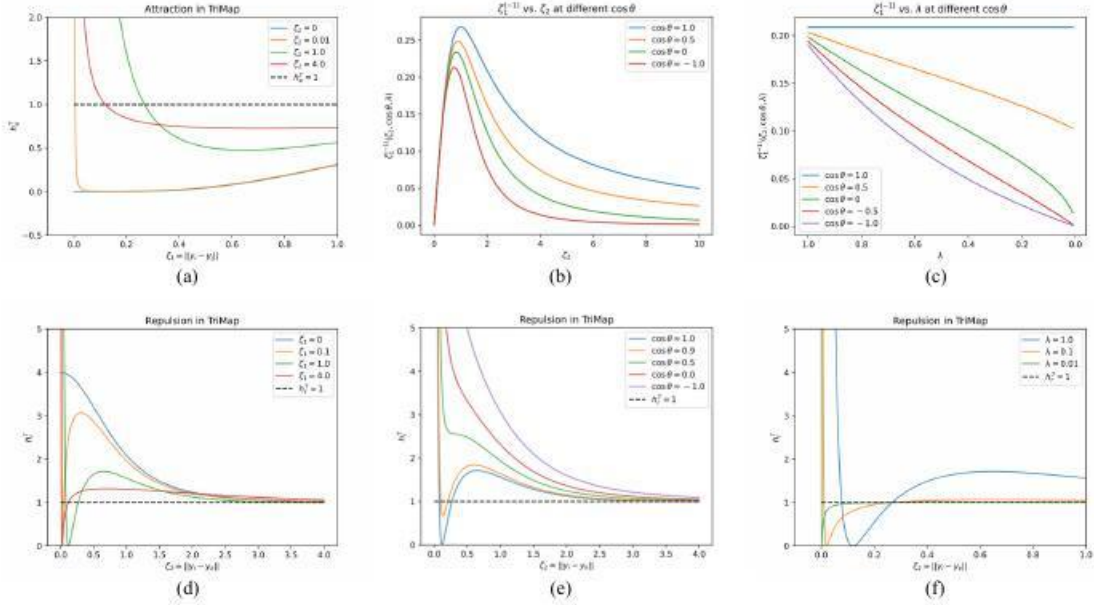


Figure 13: TriMap attraction and repulsion behavior. (a) h_a^T as a function of ζ_1 for multiple ζ_2 values; values below the dotted line indicate attraction. As in UMAP, TriMap can produce both attraction and repulsion for nearest-neighbor pairs (y_i, y_j) . (b,c) Unlike UMAP, the minimum contraction distance ($\zeta_1^{(-1)}$) depends on (b) ζ_2 and (c) λ , with $\cos \theta$ controlling the range. (d-f) TriMap repulsion behavior under variation of (d) ζ_1 , (e) $\cos \theta$, and (f) λ . Values above (below) the dotted line indicate repulsion (attraction). In contrast to UMAP repulsion, TriMap repulsion can induce either expansion or contraction. Unless otherwise labeled, $\zeta_1 = 1.0$, $\zeta_2 = 0.5$, $\cos \theta = 1.0$, and $\lambda = 1.0$.

and the repulsion shape is

$$f_r^T(\zeta_1, \zeta_2) = \frac{2(1 + \zeta_1^2)}{(2 + \zeta_1^2 + \zeta_2^2)^2}. \quad (49)$$

The TriMap attraction and repulsion shapes in Fig. 12 follow trends similar to UMAP. However, the minimum attraction value is -0.5 ; consequently, attraction alone does not flip pair positions as it can in UMAP. Because Eqs. (45)-(47) couple attractive and repulsive terms, Propositions 4.1 and 4.2 do not apply directly. We first analyze the attraction condition:

Proposition G.1. *Update equations (45)-(47) provide a contraction if*

$$h_a^T(\zeta_1, \zeta_2, \theta, \lambda) < 1, \quad (50)$$

where $h_a^T(\zeta_1, \zeta_2, \theta, \lambda) = (1 + 2\lambda f_a^T)^2 + 2(1 + 2\lambda f_a^T)\lambda f_r^T \frac{\zeta_2}{\zeta_1} \cos \theta + (\lambda f_r^T)^2 \frac{\zeta_2^2}{\zeta_1^2}$ and θ is the angle between the vectors $(y_i - y_j)$ and $(y_i - y_k)$, i.e., $\cos \theta = \frac{(y_i^t - y_j^t)^T (y_i^t - y_k^t)}{\|y_i^t - y_j^t\|_2 \|y_i^t - y_k^t\|_2}$.

Proof. The contraction requirement is

$$\|y_i^{t+1} - y_j^{t+1}\|_2^2 < \|y_i^t - y_j^t\|_2^2. \quad (51)$$

From Eq. (45) and (46), $y_i^{t+1} - y_j^{t+1} = (1 + 2\lambda f_a^T)(y_i^t - y_j^t) + \lambda f_r^T(y_i^t - y_k^t)$. Substituting this expression into Eq. (51) yields the stated inequality. $\square$

Inequality (50) depends jointly on ζ_1 , ζ_2 , $\cos \theta$, and λ . Thus the nearest-neighbor distance $\zeta_1 = \|y_i - y_j\|_2$, whose contraction is desired, is coupled to additional geometric and optimization variables. Figure 13(a) shows the attraction regime for several ζ_2 values with $\cos \theta = 1$ and $\lambda = 1$. Values below the dotted line indicate attraction and contraction of ζ_1 , whereas values above the line indicate repulsion and expansion of ζ_1 . The intersection of h_a^T with the dotted line defines the minimum contraction distance ($\zeta_1^{(-1)}$), analogous to UMAP's ζ_{-1} :

$$\zeta_1^{(-1)}(\zeta_2, \theta, \lambda) = \arg \min_{\zeta_1} |h_a^T(\zeta_1, \zeta_2, \theta, \lambda) - 1|, \quad (52)$$

$$\text{s.t. } \zeta_1 \geq 0. \quad (53)$$

For most configurations, $\zeta_1^{(-1)}$ is finite and positive (Fig. 13(b,c)). Consequently, TriMap can exhibit UMAP-like contraction-threshold behavior and may require learning-rate annealing or an analogous adaptation strategy; the TriMap implementation uses delta-bar-delta (Jacobs, 1988) under appropriate initialization.

Proposition G.2. *Update equations (45)-(47) provide expansion if*

$$h_r^T(\zeta_1, \zeta_2, \theta, \lambda) > 1, \quad (54)$$

where $h_r^T(\zeta_1, \zeta_2, \theta, \lambda) = (1 + 2\lambda f_r^T)^2 + 2\lambda f_a^T(1 + \lambda f_r^T) \frac{\zeta_1}{\zeta_2} \cos \theta + (\lambda f_a^T)^2 \frac{\zeta_1^2}{\zeta_2^2}$.

Proof. The expansion requirement is

$$\|y_i^{t+1} - y_k^{t+1}\|_2^2 > \|y_i^t - y_k^t\|_2^2. \quad (55)$$

From Eq. (45) and (47), $y_i^{t+1} - y_k^{t+1} = (1 + 2\lambda f_r^T)(y_i^t - y_k^t) + \lambda f_a^T(y_i^t - y_j^t)$. Substituting this expression into Eq. (55) gives inequality (54). $\square$

Inequality (54) also depends on ζ_1 , ζ_2 , $\cos \theta$, and λ . The relevant quantity is expansion of $\zeta_2 = \|y_i - y_k\|_2$. Figures 13(d-f) show how the repulsion regime changes as the remaining quantities vary. Values above the dotted line indicate repulsion and distance expansion, whereas values below the line indicate attraction. The key contrast with UMAP is that TriMap repulsion can produce contraction rather than expansion. Because this anomalous regime occurs at small distances, it can be mitigated through appropriate initialization and triplet selection.

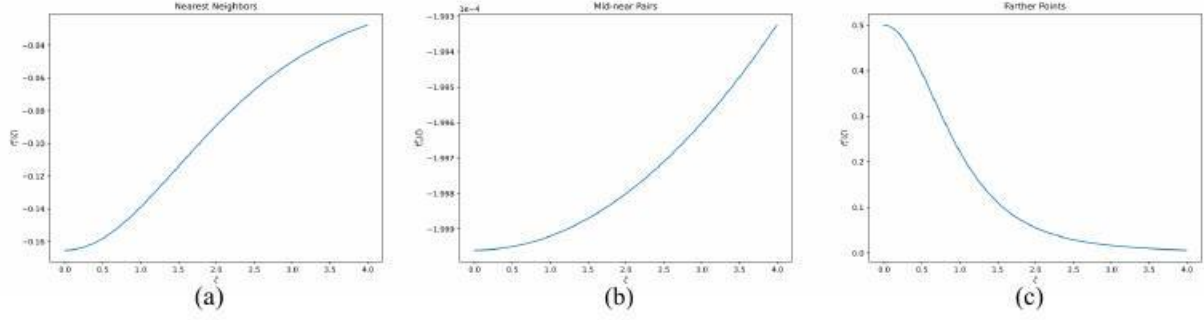


Figure 14: PaCMAP attraction and repulsion shapes. (a,b) Attraction shapes for (a) nearest-neighbor and (b) mid-near pairs; the mid-near values are on the order of 10^{-4} . (c) Repulsion shape for farther pairs. All panels use $\lambda = 1$.

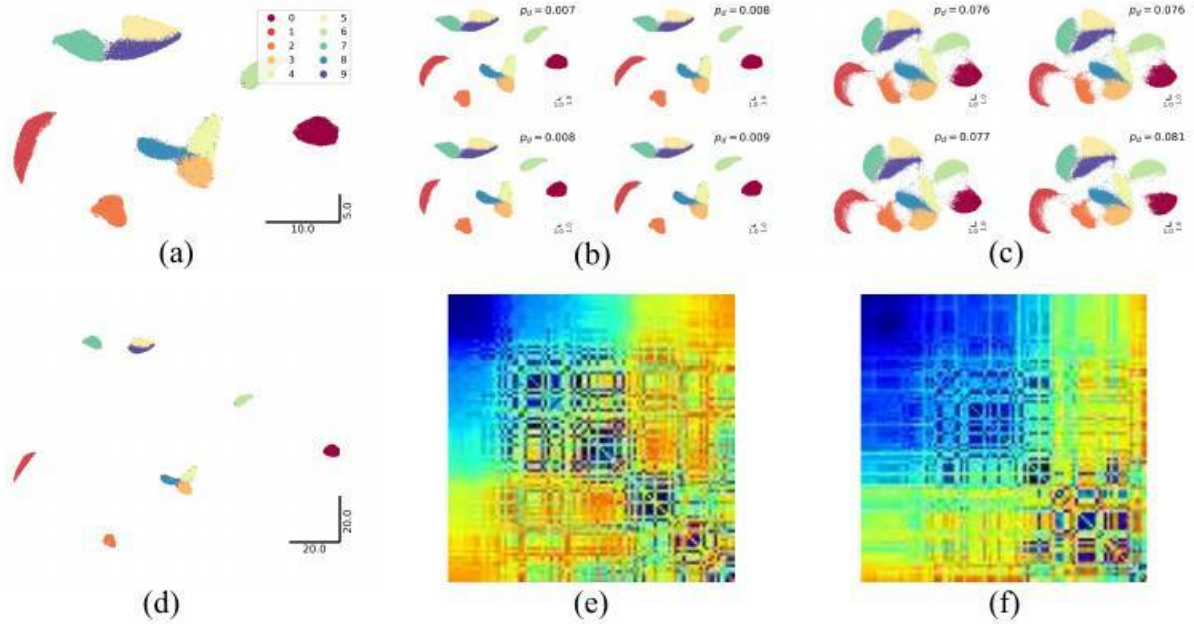


Figure 15: PaCMAP behavior under alternative shape conditions. (a) PCA-initialized PaCMAP embedding of MNIST. (b,c) Four randomly initialized samples with the closest match to (a), shown for (b) original PaCMAP and (c) PaCMAP with modified attraction shape $f_a^M(\zeta) = f_a^P(\zeta) - 0.001\zeta$. (d) MNIST PaCMAP embedding with modified repulsion shape $f_r^M = f_r^P + 0.00005$. (e,f) Procrustes matrices for (e) original PaCMAP (0.51 ± 0.22) and (f) PaCMAP with modified attraction shape (0.43 ± 0.22). As in UMAP, stronger long-distance attraction improves consistency, whereas stronger repulsion produces smaller clusters and larger inter-cluster distances.

G.2 Pairwise Controlled Manifold Approximation (PaCMAP)

PaCMAP (Wang et al. 2021) optimizes the low-dimensional embedding using interactions at multiple scales. Its loss is

$$\mathcal{L}^P = w_{NB} \sum_{(i,j) \in NN} \frac{1}{1 + 10q_{ij}} + w_{MN} \sum_{(i,k) \in MN} \frac{1}{1 + 10000q_{ik}} + w_{FP} \sum_{(i,l) \in FP} \frac{1}{1 + \frac{1}{q_{il}}}, \quad (56)$$

where w_{NB} , w_{MN} , and w_{FP} weight nearest-neighbor (NN), mid-near (MN), and further-pair (FP) interactions, respectively (details in Appendix H). The first two terms are attractive, and the final term is repulsive. The objective can be interpreted as a modified TriMap triplet loss in which the affinities for distant points in the attractive terms are replaced by constants ($1/10$ for nearest neighbors, $1/10000$ for mid-near pairs, and 1 for farther pairs). This modification separates the loss into three components and decouples the update equations. The nearest-neighbor updates

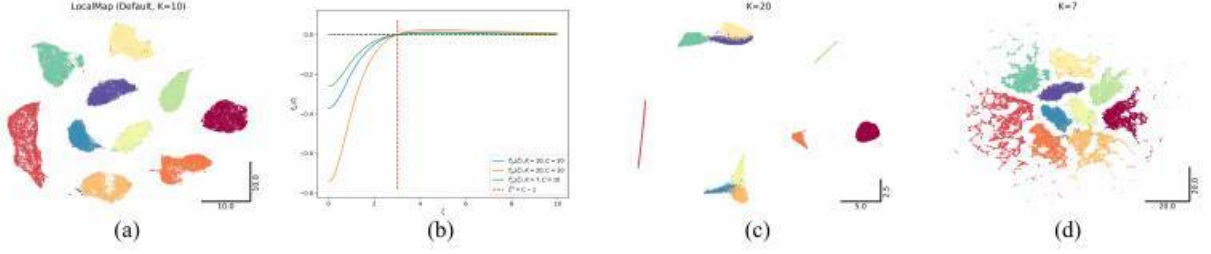


Figure 16: LocalMAP behavior on MNIST. (a) Default embedding. (b) Nearest-neighbor attraction–repulsion shape; values below (above) the dotted line indicate attraction (repulsion). The transition from attraction to repulsion occurs when ζ crosses $\sqrt{C-1}$. Following the LocalMAP implementation, we use $C = 10$. (c) For large $K (= 20)$, repulsion dominates and clusters become compact. (d) For small $K (= 7)$, attraction dominates and existing clusters split.

are

$$y_i^{t+1} = y_i^t + \lambda f_a^P(\zeta_1^t)(y_i^t - y_j^t), \quad (57)$$

$$y_j^{t+1} = y_j^t - \lambda f_a^P(\zeta_1^t)(y_i^t - y_j^t), \quad (58)$$

the mid-near-pair updates are

$$y_i^{t+1} = y_i^t + \lambda f_m^P(\zeta_2^t)(y_i^t - y_k^t), \quad (59)$$

$$y_k^{t+1} = y_k^t - \lambda f_m^P(\zeta_2^t)(y_i^t - y_k^t), \quad (60)$$

and the farthest-pair updates are

$$y_i^{t+1} = y_i^t + \lambda f_r^P(\zeta_3^t)(y_i^t - y_l^t), \quad (61)$$

$$y_l^{t+1} = y_l^t - \lambda f_r^P(\zeta_3^t)(y_i^t - y_l^t), \quad (62)$$

where $\zeta_1 = \|y_i - y_j\|_2$, $\zeta_2 = \|y_i - y_k\|_2$, and $\zeta_3 = \|y_i - y_l\|_2$ are the relevant distances. The functions f_a^P and f_m^P are the attraction shapes for nearest-neighbor and mid-near pairs, respectively, and f_r^P is the repulsion shape for farthest pairs. Their functional forms are

$$f_a^P(\zeta) = -\frac{20}{(11 + \zeta^2)^2}, \quad (63)$$

$$f_m^P(\zeta) = -\frac{20000}{(10001 + \zeta^2)^2}, \quad (64)$$

$$f_r^P(\zeta) = \frac{2}{(2 + \zeta^2)^2}. \quad (65)$$

f_a^P and f_m^P satisfy Proposition 4.1 and f_r satisfies Proposition 4.2 (Fig. 14). PaCMAP attraction is weaker than UMAP attraction, but it remains effective over a wide learning-rate range when combined with Adam optimization (Kingma and Ba, 2015); with $w_{NB} = 3$, the maximum attraction magnitude remains below 0.5, preventing pair flips during attractive updates. PaCMAP typically initializes the embedding inside a small low-dimensional sphere, often on the order of 10^{-3} in (Wang et al., 2021), and then relies on repulsion to separate clusters. Overall, PaCMAP recovers most UMAP clustering properties, especially on MNIST, while improving cluster ordering. Its random-initialization consistency is better than UMAP’s (Fig. 15(b,e)) and can be improved further with modified attraction (Fig. 15(c,f)). Adding a small constant to the repulsion shape increases embedding compactness by increasing inter-cluster distance.

G.3 Pairwise Controlled Manifold Approximation with Local Adjusted Graph (LocalMAP)

LocalMAP (Wang et al., 2025) extends PaCMAP. A defining empirical property of LocalMAP is that it separates all 10 MNIST clusters (Fig. 16(a)), exhibiting behavior similar to Figs. 3(g,h,j,k) and Figs. 4(a,b). Here we analyze how its attractive and repulsive components regulate cluster compactness and connectedness.

LocalMAP first applies PaCMAP and then performs an additional optimization stage that modifies nearest-neighbor interactions to decouple selected clusters. The corresponding loss is

$$\mathcal{L}^L = \sum_{(i,j) \in NN} \frac{K}{\frac{1}{\sqrt{q_{ij}}} + C\sqrt{q_{ij}}} + \sum_{(i,l) \in FP} \frac{1}{1 + \frac{1}{q_{il}}}. \quad (66)$$

The first term combines attractive and repulsive behavior for the same nearest-neighbor pair. In one distance regime it induces attraction, and in the other it induces repulsion. The second term matches the PaCMAP farther-pair term, except that LocalMAP resamples farther pairs every few iterations. We therefore analyze only the nearest-neighbor term. The update equations are

$$y_i^{t+1} = y_i^t + \lambda f_{ar}^L(\zeta_i^t)(y_i^t - y_j^t), \quad (67)$$

$$y_j^{t+1} = y_j^t - \lambda f_{ar}^L(\zeta_j^t)(y_i^t - y_j^t), \quad (68)$$

where f_{ar}^L denotes the attraction–repulsion shape

$$f_{ar}^L(\zeta) = -\frac{K(C-1-\zeta^2)}{2\sqrt{1+\zeta^2}(1+C+\zeta^2)^2}. \quad (69)$$

The update contracts distances when $\zeta^2 < C-1$ and $-1 < \lambda f_{ar}^L < 0$, by Proposition 4.1. When $\zeta^2 > C-1$, $f_{ar} > 0$, and Proposition 4.2 implies expansion. The parameters K and C determine which regime dominates the dynamics. The LocalMAP implementation sets both C and K to 10 (Fig. 16b), yielding attraction that is stronger than PaCMAP attraction (Fig. 14a). Consequently, nearest-neighbor pairs with $\zeta > 3$ experience repulsion, which separates pairs that bridge two clusters. The parameter K acts as a force-scaling coefficient and regulates this separation.

Using Proposition 4.1 the condition $\lambda f_{ar}^L(0) \geq -\frac{1}{2}$ gives the largest admissible K and the strongest attraction allowed by f_{ar}^L without flipping pair placements. Avoiding flips during the LocalMAP stage helps preserve the PaCMAP ordering; otherwise, cluster separation may be inhibited. This condition simplifies Eq. (69) to $K \leq \frac{\lambda(1+C)^2}{C-1}$, which gives $K \lesssim 13.44$ for $C = 10$ and $\lambda = 1$. For larger $K \gtrsim 13.44$, repulsion dominates and clusters become more compact, but the LocalMAP objective is not achieved because the bridge between clusters persists for $K = 20$ (Fig. 16c). For smaller K , attraction dominates because repulsion is too weak, and existing clusters split (Fig. 16d) for $K = 7$, matching the behavior in Fig. 4b).

G.4 t -distributed Stochastic Neighbor Embedding (t -SNE)

t -SNE (Van der Maaten and Hinton 2008) optimizes a pairwise divergence between high- and low-dimensional affinities. The loss is

$$\mathcal{L} = -\sum_{i,j} w_{i,j} \log \left(\frac{q_{i,j}}{\sum_{k \neq l} q_{k,l}} \right), \quad (70)$$

where $w_{i,j}$ is the pair weight. The original t -SNE implementation includes all pairs rather than only nearest neighbors. The loss decomposes into attractive and repulsive components as

$$\mathcal{L} = \sum_{i,j} \left[-w_{i,j} \log(q_{i,j}) + w_{i,j} \log \left(\sum_{k \neq l} q_{k,l} \right) \right]. \quad (71)$$

As above, the first term induces attraction and the second term induces repulsion. The attractive term matches UMAP with $a = 1$ and $b = 1$ and is straightforward to compute, whereas the repulsive term couples all pairs and is therefore computationally expensive. Applying the same decomposition used for UMAP gives the t -SNE update equations. Because original t -SNE does not use a nearest-neighbor graph, the all-pairs weights $w_{i,j}$ are central to the updates. The attractive updates are

$$y_i^{t+1} = y_i^t + \lambda w_{i,j} f_a^{t-SNE}(\zeta_{i,j}^t)(y_i^t - y_j^t), \quad (72)$$

$$y_j^{t+1} = y_j^t - \lambda w_{i,j} f_a^{t-SNE}(\zeta_{i,j}^t)(y_i^t - y_j^t), \quad (73)$$

where f_a^{t-SNE} is the t -SNE attraction shape and $\zeta_{i,j} = \|y_i - y_j\|_2$. The repulsive updates are

$$y_i^{t+1} = y_i^t + \lambda \frac{w_{i,j}}{Z} \sum_k f_r^{t-SNE}(\zeta_{i,k}^t)(y_i^t - y_k^t), \quad (74)$$

$$y_j^{t+1} = y_j^t - \lambda \frac{w_{i,j}}{Z} \sum_l f_r^{t-SNE}(\zeta_{l,j}^t)(y_l^t - y_j^t), \forall j, j \neq i, \quad (75)$$

where f_r^{t-SNE} is the t -SNE repulsion shape and $Z = \sum_{k \neq l} q_{k,l}$. The shape functions are

$$f_a^{t-SNE}(\zeta) = -\frac{2}{1+\zeta^2}, \text{ and} \quad (76)$$

$$f_r^{t-SNE}(\zeta) = \frac{2}{(1+\zeta^2)^2}. \quad (77)$$

From the attractive updates in Eqs. (72)-(73), f_a^{t-SNE} satisfies Proposition 4.1 under the corresponding effective coefficient condition and yields a minimum contraction distance ζ_{-1} . The repulsive update is coupled across all pairs, so it does not reduce to a simple one-pair repulsion-shape condition. Using the TriMap analysis as a template, we derive the following result:

Proposition G.3. *The update Eqs. (74)-(75) provide an expansion if*

$$h_r^{t-SNE}(\zeta_{i,j}, v, \theta, \lambda, w_{i,j}) > 1, \quad (78)$$

where

$$\begin{aligned} h_r^{t-SNE}(\zeta_{i,j}, v, \theta, \lambda, w_{i,j}) &= (1 + 2\lambda \frac{w_{i,j}}{Z} f_r^{t-SNE}(\zeta_{i,j}))^2 + \frac{\|v\|_2^2}{\zeta_{i,j}^2} + 2\lambda \frac{w_{i,j}}{Z} f_r^{t-SNE}(\zeta_{i,j}) \frac{\|v\|_2}{\zeta_{i,j}} \cos \theta, \\ v &= \lambda \frac{w_{i,j}}{Z} \left(\sum_{k,k \neq j} f_r^{t-SNE}(\zeta_{i,k})(y_i - y_k) + \sum_{l,l \neq i} f_r^{t-SNE}(\zeta_{l,j})(y_l - y_j) \right), \\ \text{and } \theta &\text{ is the angle between } (y_i - y_j) \text{ and } v. \end{aligned}$$

Proof. The expansion condition is

$$\|y_i^{t+1} - y_j^{t+1}\|_2^2 > \|y_i^t - y_j^t\|_2^2. \quad (79)$$

From Eqs. (74) and (75), $y_i - y_j = (1 + 2\lambda \frac{w_{i,j}}{Z} f_r^{t-SNE}(\zeta_{i,j}))(y_i - y_j) + v$. Substituting this expression into Eq. (79) gives the stated inequality. $\square$

Because the t -SNE repulsion condition in Eq. (78) has the same structure as the TriMap condition, its repulsive update can also induce either expansion or contraction.

Because t -SNE scales its forces by $w_{i,j}$ for attraction and by $\frac{w_{i,j}}{Z}$ for repulsion, its effective attractive and repulsive magnitudes are typically smaller than UMAP's. Consequently, implementations often use large learning rates, for example approximately 10^3 in Open- t -SNE (Poličar et al. 2024). Most t -SNE implementations also include an early-exaggeration phase in which attractive forces are multiplied by a constant during the first iterations. This phase pulls pairs that should be close but are currently far apart toward one another, thereby inducing far-sightedness. It also separates points that are too close ($\zeta_{i,j} \approx \zeta_{-1}$) but should be separated. Thus, early exaggeration is a central mechanism for obtaining consistent t -SNE embeddings, especially under random initialization. When applied throughout optimization, the same mechanism makes t -SNE embeddings visually closer to UMAP embeddings (Böhm et al. 2022).

G.5 Stochastic Neighbor Embedding (SNE)

SNE (Hinton and Roweis 2002) is one of the earliest algorithms in this class. It uses the same objective form as t -SNE (Eq. 71) but sets $q_{i,j} = \exp(-\|y_i - y_j\|_2^2)$. The resulting attraction shape is

$$f_a^{SNE}(\zeta) = -2, \quad (80)$$

which satisfies Proposition 4.1 under the corresponding effective coefficient condition, together with the repulsion shape

$$f_r^{SNE}(\zeta) = 2 \exp(-\zeta^2), \quad (81)$$

which satisfies proposition G.3 after replacing the t -SNE symbols with their SNE counterparts. The attraction shape is poorly conditioned because contraction depends mainly on the learning rate (λ) and weights ($w_{i,j}$). This dependence causes clusters to overlap even with relatively few samples, a phenomenon known as the crowding problem (Van der Maaten and Hinton, 2008). t -SNE and later algorithms mitigate this issue by replacing the Gaussian kernel with a heavy-tailed kernel, such as Eq. (43).

G.6 Multidimensional Scaling (MDS)

Multidimensional scaling (MDS) (Borg and Groenen 2007) usually relies on stress majorization rather than gradient descent because direct gradient methods often fail to converge to high-quality mappings. Nevertheless, gradient-based formulations have been studied (Kruskal 1964, Zheng et al. 2018). We briefly analyze this setting here. In

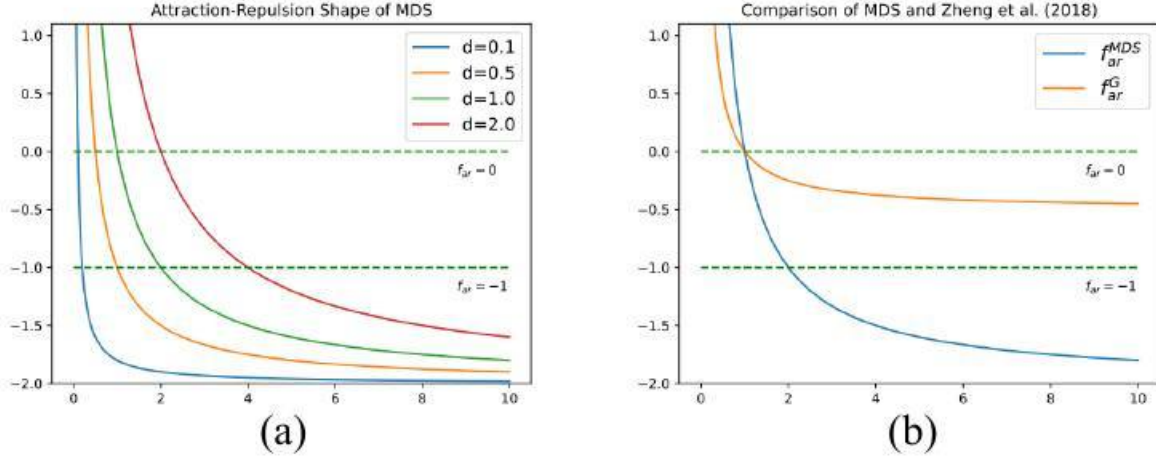


Figure 17: MDS attraction-repulsion characteristics. (a) Characteristic induced by f_{ar} . (b) Comparison between the MDS characteristic and the method of [Zheng et al. \(2018\)](#) for $d = 1.0$.

particular, [Zheng et al. \(2018\)](#) proposes a successful gradient-descent MDS variant for graph drawing. We begin with the cost function, absorbing the weight $w_{i,j}$ into the learning rate without loss of generality for this discussion:

$$\mathcal{L}^{MDS} = \sum_{i,j,i < j} (\|y_i - y_j\| - d_{ij})^2. \quad (82)$$

The loss contains a single term for each pair and does not explicitly separate attraction from repulsion; instead, each pairwise term can induce either behavior. Because the loss is separable over (i, j) components, it suffices to analyze one pair. The derivative is

$$\frac{\partial}{\partial y_i} (\|y_i - y_j\| - d_{ij})^2 = 2 \frac{\|y_i - y_j\| - d_{ij}}{\|y_i - y_j\|} (y_i - y_j). \quad (83)$$

The resulting MDS attraction-repulsion shape is

$$f_{ar}^{MDS}(\zeta) = -2 \frac{\zeta - d}{\zeta}, \quad (84)$$

where $\zeta = \|y_i - y_j\|_2$. The shape is attractive for $\zeta > d$ and is bounded in $[-2, 0]$ on that interval (Fig. 17(a)). This range is problematic for convergence because only values in $[-1, 0]$ contract; the remaining values expand. Values below -2 additionally flip and expand the pair. To avoid this behavior, [Zheng et al. \(2018\)](#) uses an ad hoc force-directed-graph-inspired formulation:

$$f_{ar}^G(\zeta) = -\frac{1}{2} \frac{\zeta - d}{\zeta}, \quad (85)$$

which is stable for effective learning rates $\lambda \leq 1$. Because f_{ar}^G is bounded in $[-0.5, 0]$, this learning-rate range preserves contraction. For $\zeta < d$, however, $f_{ar}^{MDS} > 0$ and the pairwise term is repulsive. Overall, this attraction-repulsion structure is most effective when the initialization is already close to the desired configuration. Continued optimization of a pair causes oscillation around $\zeta = d$, where $f_{ar}^G = 0$; this distance plays a role analogous to UMAP’s ζ_{-1} . No learning-rate choice drives this distance to zero, because matching the target distance is the MDS objective. Consequently, MDS clusters are often more diffuse than those produced by UMAP or PaCMAP; see [Lambert et al. \(2022\)](#), [de Bodt et al. \(2025\)](#), and [Kury et al. \(2025\)](#) for illustrations. Finally, [Lambert et al. \(2022\)](#) reformulates MDS as quartet stress, using a four-sample loss and a relative distance normalized by the sum of six quartet distances, which provides global regularization and lower computational cost than majorization.

H Construction of the High-Dimensional Graph

Stochastic Neighbor Embedding (SNE) [Hinton and Roweis \(2002\)](#) is a foundation for modern dimensionality-reduction algorithms. It constructs a high-dimensional graph over $X = \{x_i \in \mathbb{R}^n | i = 1, \dots, N\}$ using

$$w_{ij} = \frac{p_{j|i} + p_{i|j}}{2N}, \quad (86)$$

$$w_{j|i} = \frac{\exp(-\|x_i - x_j\|_2^2 / 2\sigma_i^2)}{\sum_{t \neq v} \exp(-\|x_t - x_v\|_2^2 / 2\sigma_t^2)}, \quad (87)$$

where σ_i^2 is selected so that the conditional distribution matches the user-specified *perplexity* P , defined by

$$P = 2^{H_i} \quad (88)$$

$$H_i = \sum_{j \neq i} w_{j|i} \log_2 w_{j|i}. \quad (89)$$

UMAP instead constructs its high-dimensional graph from the k-nearest-neighbor (k-NN) graph via

$$p_{i|j} = \begin{cases} \exp\left(-\frac{d(x_i, x_j) - \rho_i}{\sigma_i}\right) & \text{if } x_j \in \text{KNN}(x_i, k) \\ 0 & \text{otherwise} \end{cases}, \quad (90)$$

$$\rho_i = \min_{x_j \in \text{KNN}(x_i, k)} d(x_i, x_j), \quad (91)$$

where $\text{KNN}(x_i, k)$ denotes the set of k nearest neighbors of x_i , and σ_i is a scaling parameter chosen so that $\sum_j p_{i|j} = \log_2(k)$. The directed graph is then symmetrized using a t-conorm:

$$p_{i,j} = p_{i|j} + p_{j|i} - p_{i|j}p_{j|i}. \quad (92)$$

PaCMAP uses the k-NN graph directly for affinities, assigning all such affinities value 1, together with a self-tuning distance measure [Zelnik-Manor and Perona 2004](#):

$$d_{i,j}^2 = \frac{\|x_i - x_j\|^2}{\sigma_i \sigma_j}, \quad (93)$$

where σ_i is the average Euclidean distance from x_i to its fourth, fifth, and sixth nearest neighbors. This parameter serves the same normalization role as the corresponding σ_i terms in Eq. (87) and Eq. (90), although the three definitions differ.

Despite these differences in graph construction, the subsequent optimization is governed primarily by the attraction and repulsion shapes identified in our analysis.

I Detailed Results for Varying λ_a and λ_b

This section reports the full results for the experiments summarized in Fig. 1(h,i). The results are obtained by replacing UMAP’s attraction and repulsion shapes with the corresponding shapes from the comparison methods. Figure 18 reproduces the main-text summary. Figures 19-22 show the individual embeddings for each λ_a and λ_r setting, together with the initialization, a reference embedding obtained with annealed learning rate λ , and the degenerate cases in which either λ_a or λ_r is set to 0. For PaCMAP, which includes mid-near pairs, an additional reference embedding with mid-near pairs included is also shown. Figures 23-27 and Figures 28-32 report the same protocol on FMNIST and single-cell transcriptomes, respectively.

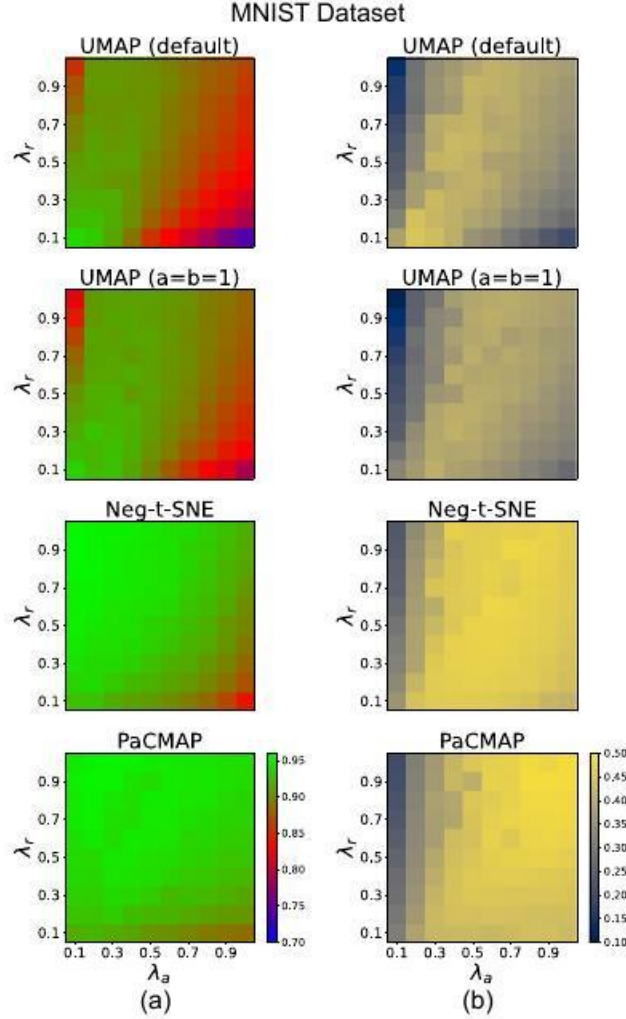


Figure 18: MNIST evaluation metrics for the compared methods: (a) trustworthiness and (b) silhouette score.

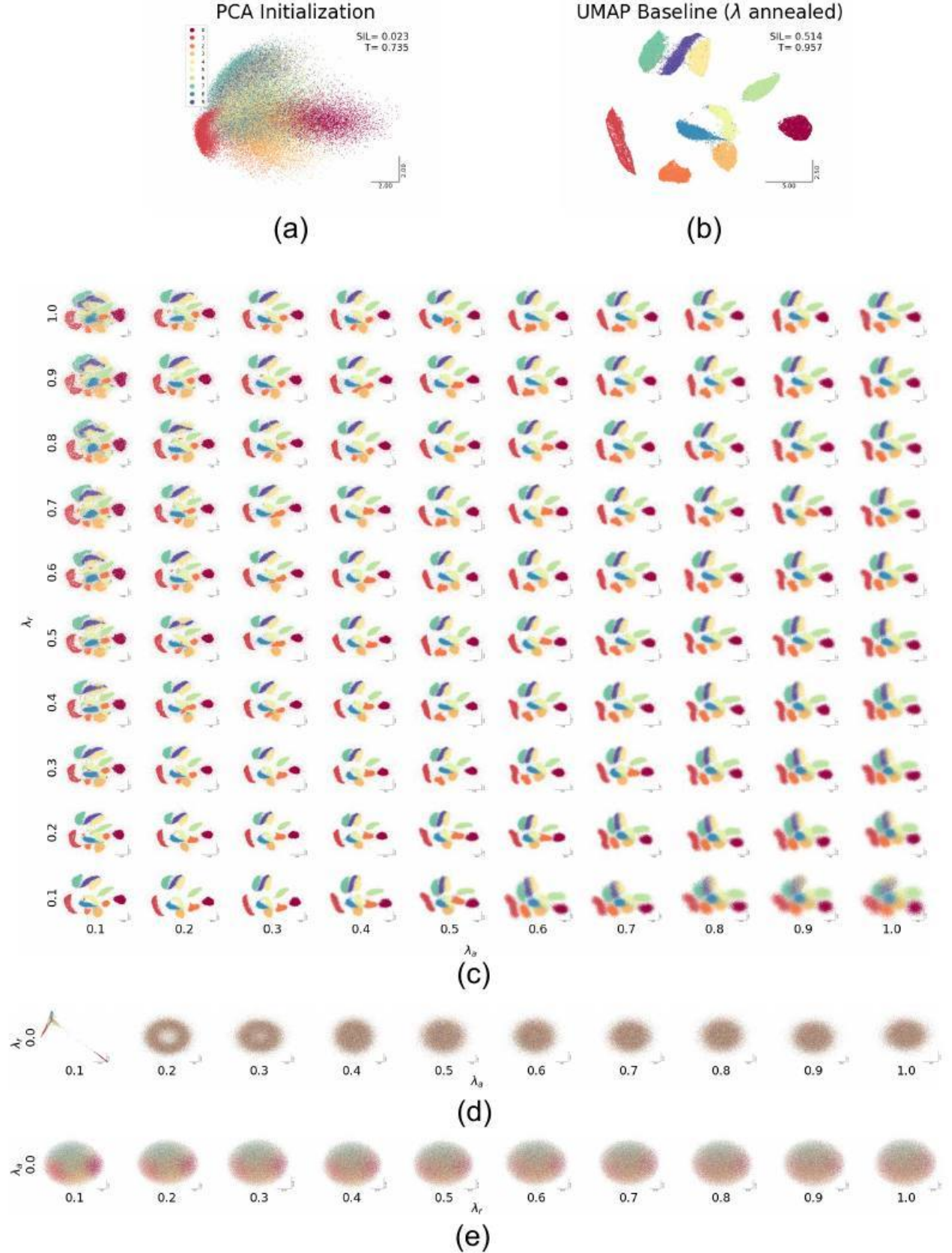


Figure 19: Learning-rate sweep over λ_a and λ_r on MNIST using UMAP's attraction and repulsion shapes. (a) Embedding initialization. (b) Baseline obtained by annealing λ from 1. (c) Embeddings obtained when λ_a and λ_r are varied without annealing. (d) When λ_r is set to 0 (no repulsion), attraction alone does not produce clusters. (e) Similarly, when λ_a is set to 0 (no attraction), repulsion alone does not produce clusters.

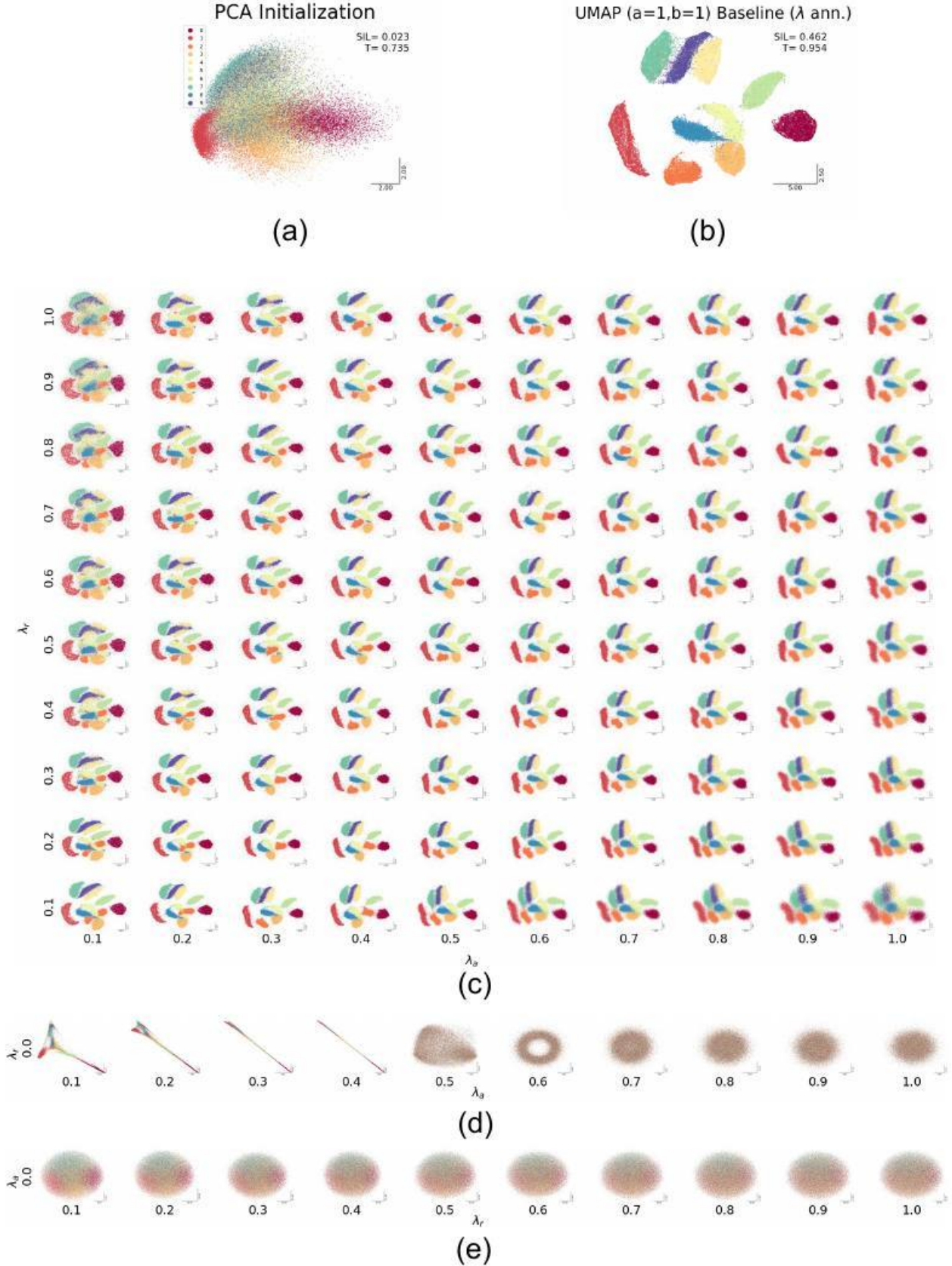


Figure 20: Learning-rate sweep over λ_a and λ_r on MNIST using UMAP's attraction and repulsion shapes (by setting $a = 1$ and $b = 1$). (a) Embedding initialization. (b) Baseline obtained by annealing λ from 1. (c) Embeddings obtained when λ_a and λ_r are varied without annealing. (d) When λ_r is set to 0 (no repulsion), attraction alone does not produce clusters. (e) Similarly, when λ_a is set to 0 (no attraction), repulsion alone does not produce clusters.

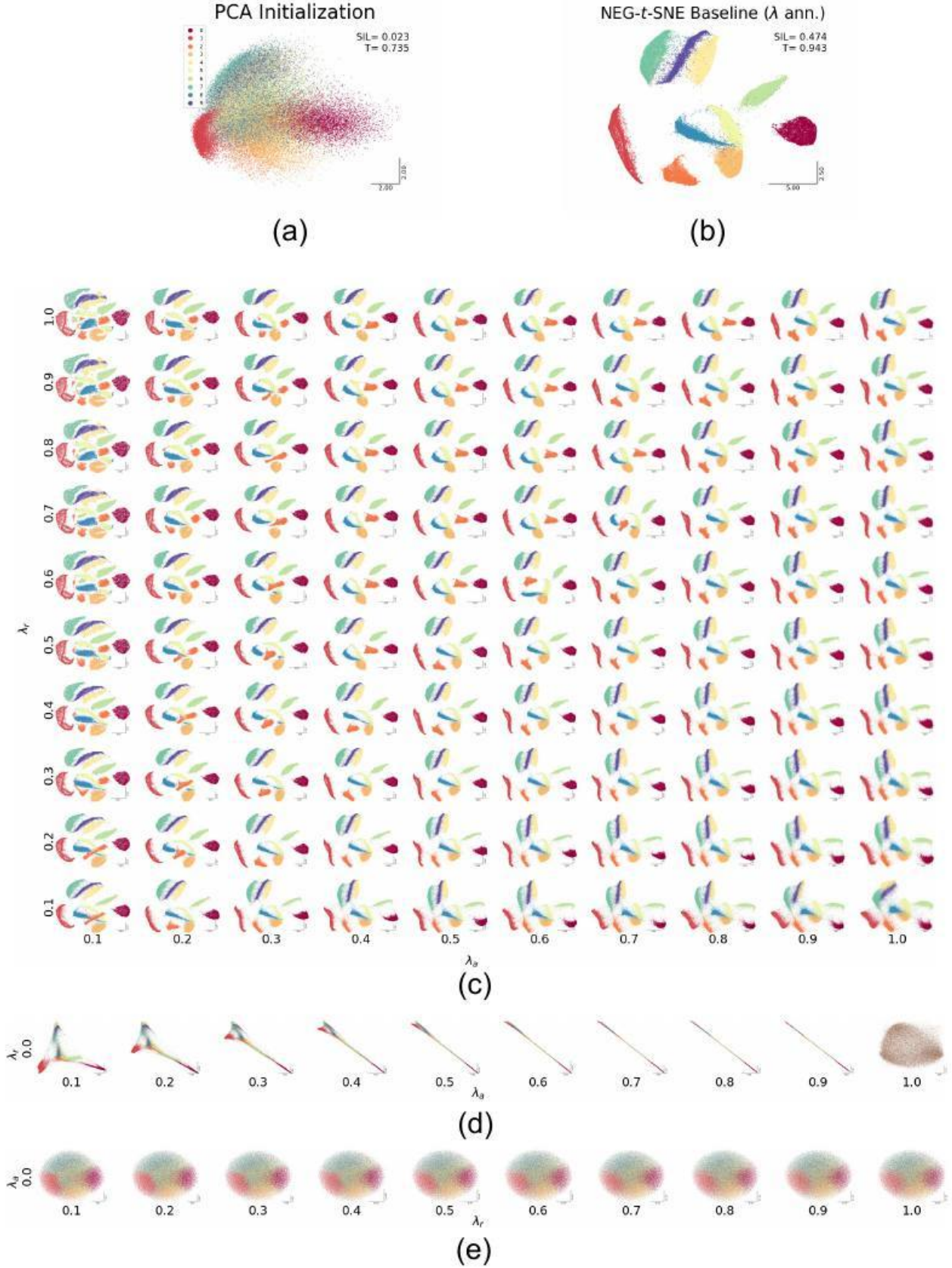


Figure 21: Learning-rate sweep over λ_a and λ_r on MNIST using NEG-t-SNE's attraction and repulsion shapes. (a) Embedding initialization. (b) Baseline obtained by annealing λ from 1. (c) Embeddings obtained when λ_a and λ_r are varied without annealing. (d) When λ_r is set to 0 (no repulsion), attraction alone does not produce clusters. (e) Similarly, when λ_a is set to 0 (no attraction), repulsion alone does not produce clusters.

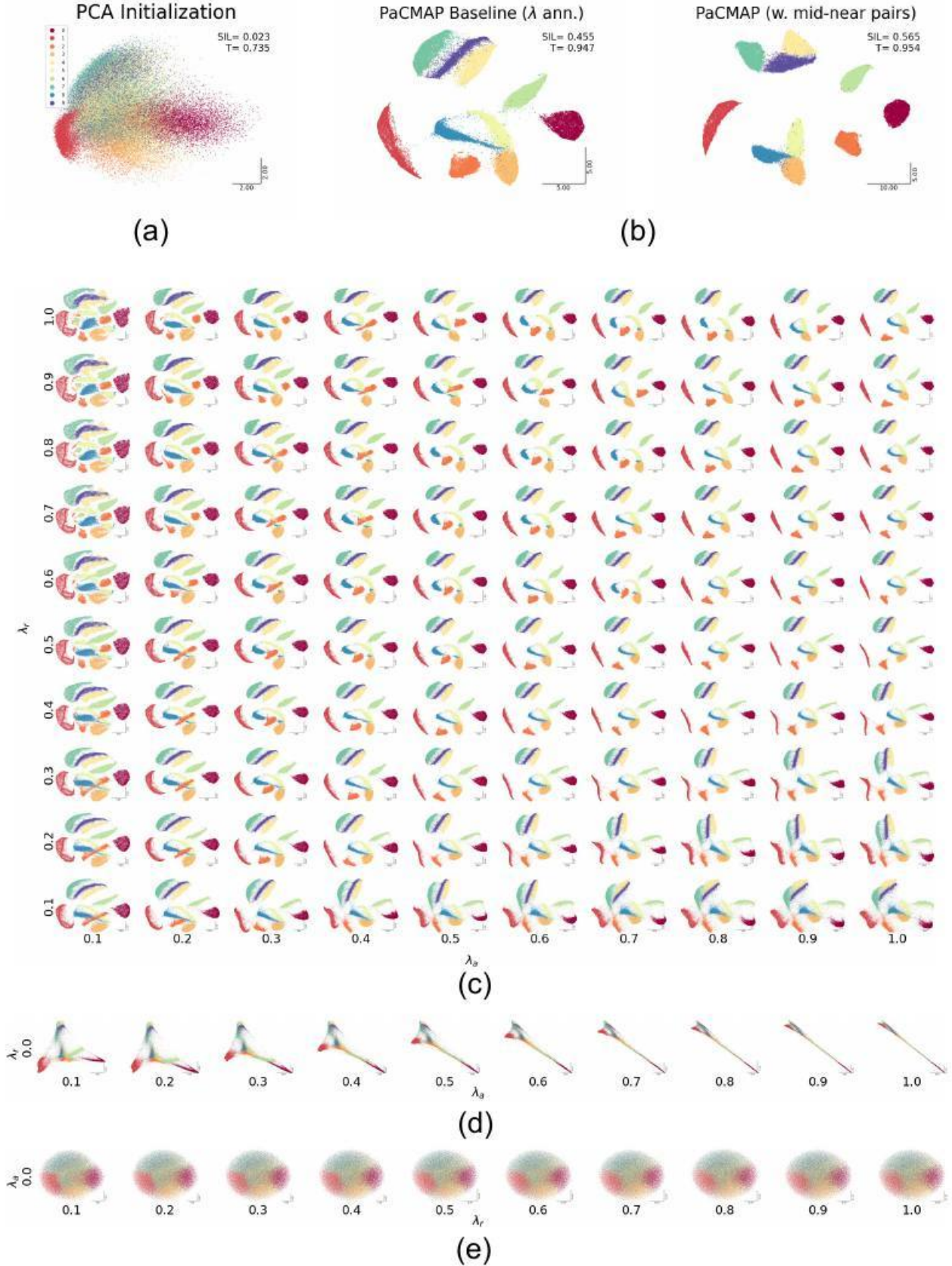


Figure 22: Learning-rate sweep over λ_a and λ_r on MNIST using PaCMAP’s attraction and repulsion shapes. (a) Embedding initialization. (b) Left: baseline obtained by annealing λ from 1 with mid-near points excluded to isolate attraction–repulsion interaction; right: baseline with mid-near points included. (c) Embeddings obtained when λ_a and λ_r are varied without annealing. (d) When λ_r is set to 0 (no repulsion), attraction alone does not produce clusters. (e) Similarly, when λ_a is set to 0 (no attraction), repulsion alone does not produce clusters.

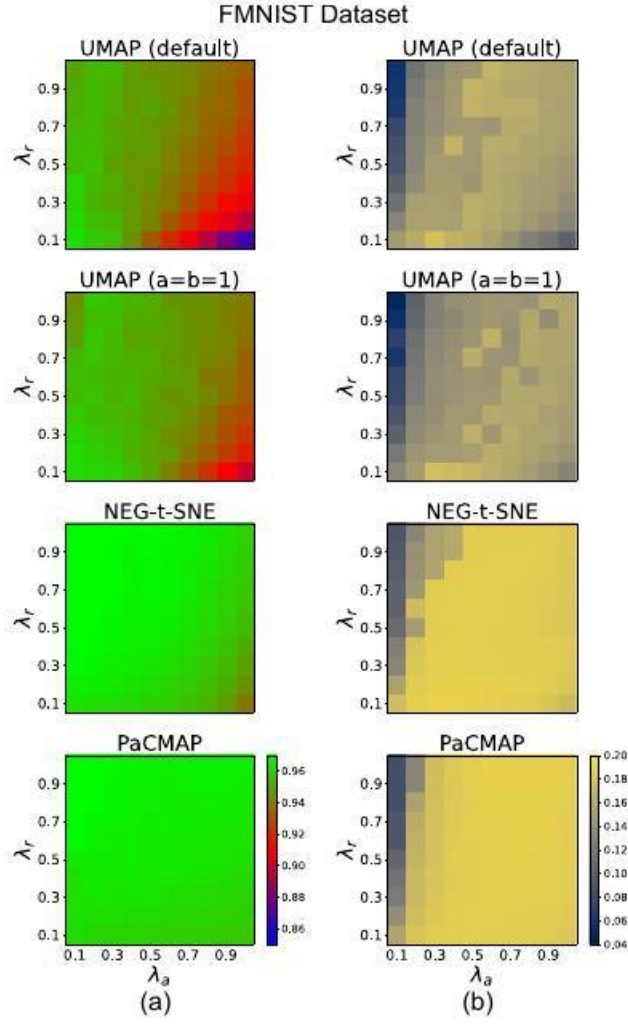


Figure 23: FMNIST evaluation metrics for the compared methods: (a) trustworthiness and (b) silhouette score.

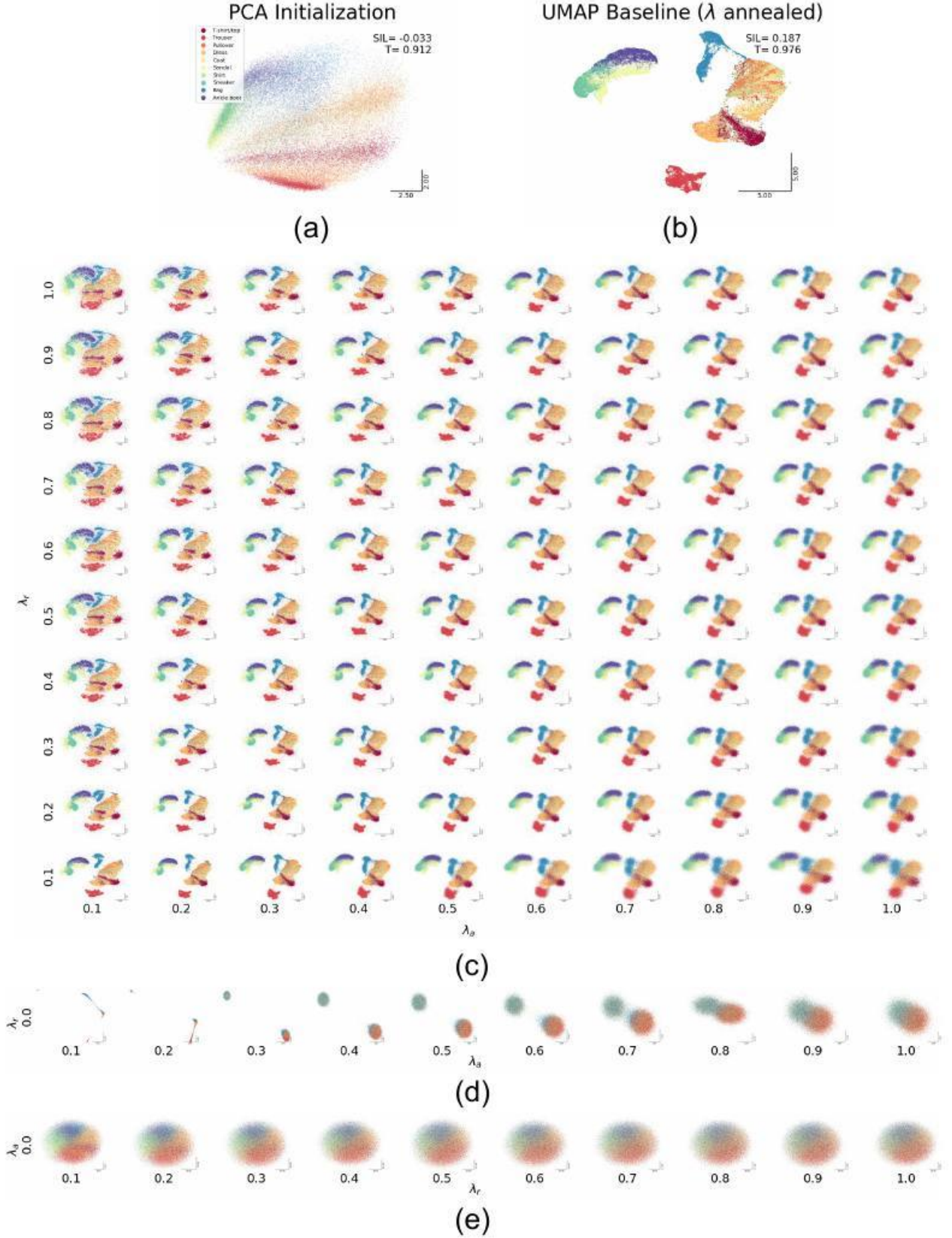


Figure 24: Learning-rate sweep over λ_a and λ_r on FMNIST using UMAP’s attraction and repulsion shapes. (a) Embedding initialization. (b) Baseline obtained by annealing λ from 1. (c) Embeddings obtained when λ_a and λ_r are varied without annealing. (d) When λ_r is set to 0 (no repulsion), attraction alone does not produce clusters. (e) Similarly, when λ_a is set to 0 (no attraction), repulsion alone does not produce clusters.

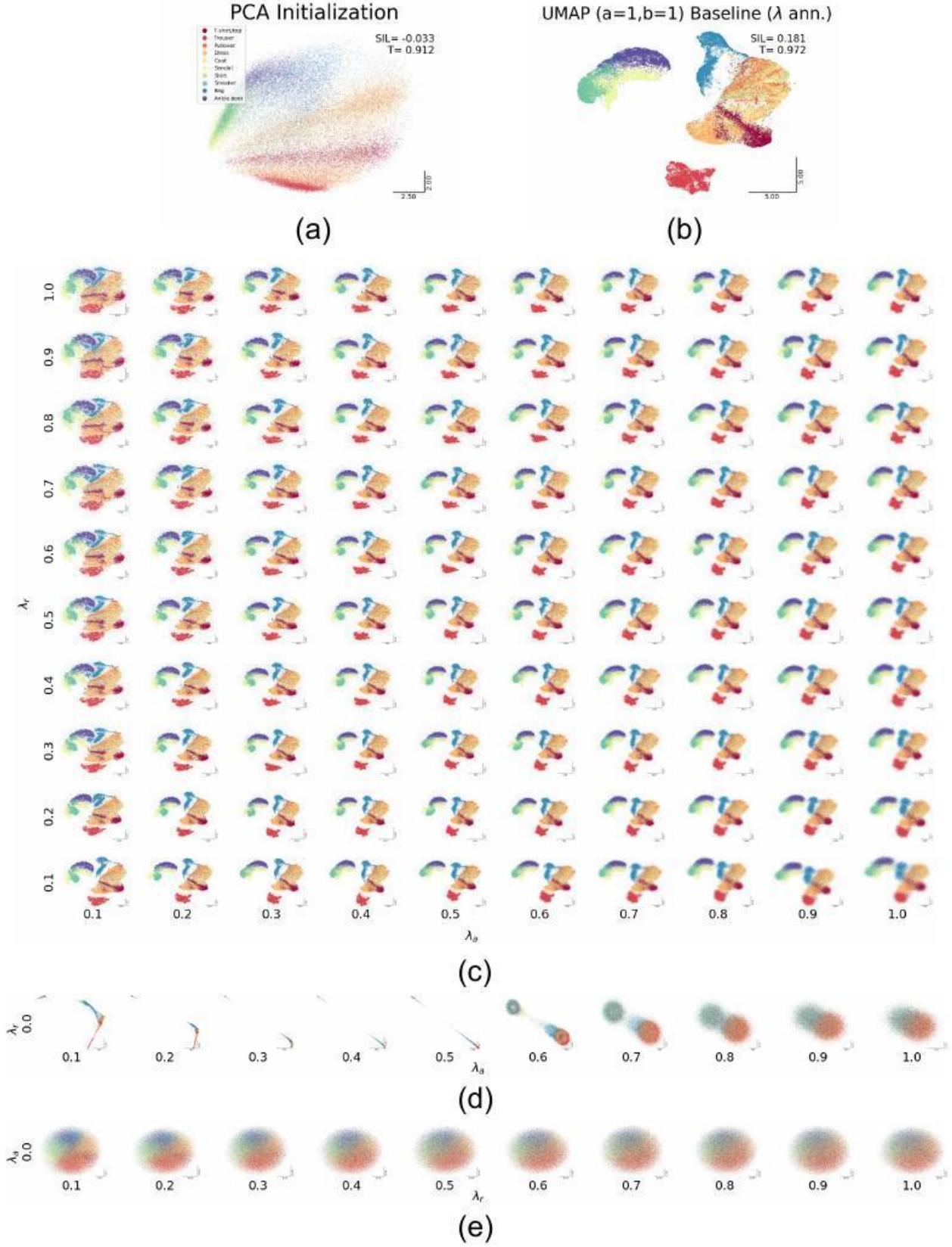


Figure 25: Learning-rate sweep over λ_a and λ_r on FMNIST using UMAP's attraction and repulsion shapes (by setting $a = 1$ and $b = 1$). (a) Embedding initialization. (b) Baseline obtained by annealing λ from 1. (c) Embeddings obtained when λ_a and λ_r are varied without annealing. (d) When λ_r is set to 0 (no repulsion), attraction alone does not produce clusters. (e) Similarly, when λ_a is set to 0 (no attraction), repulsion alone does not produce clusters.

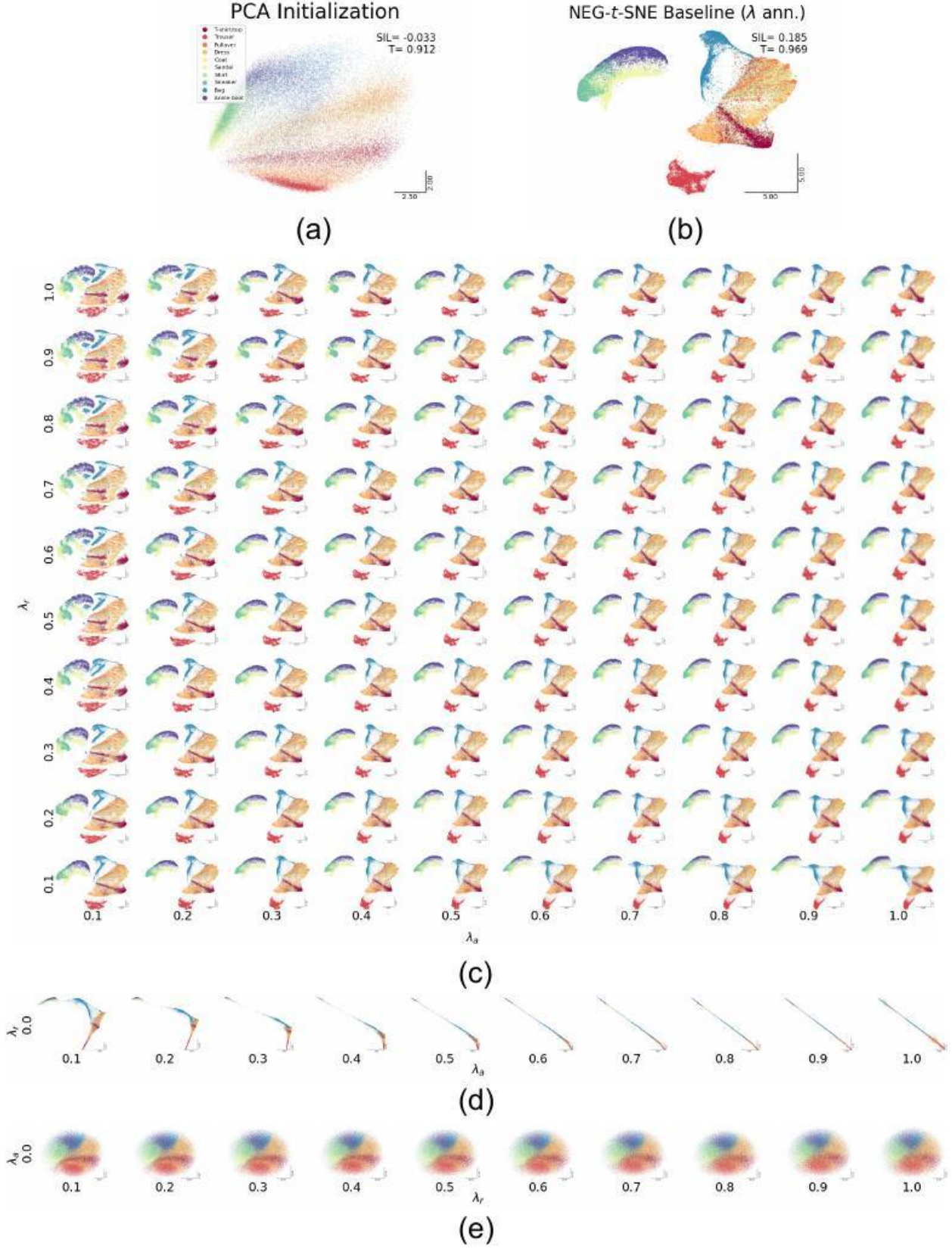


Figure 26: Learning-rate sweep over λ_a and λ_r on FMNIST using NEG-t-SNE's attraction and repulsion shapes. (a) Embedding initialization. (b) Baseline obtained by annealing λ from 1. (c) Embeddings obtained when λ_a and λ_r are varied without annealing. (d) When λ_r is set to 0 (no repulsion), attraction alone does not produce clusters. (e) Similarly, when λ_a is set to 0 (no attraction), repulsion alone does not produce clusters.

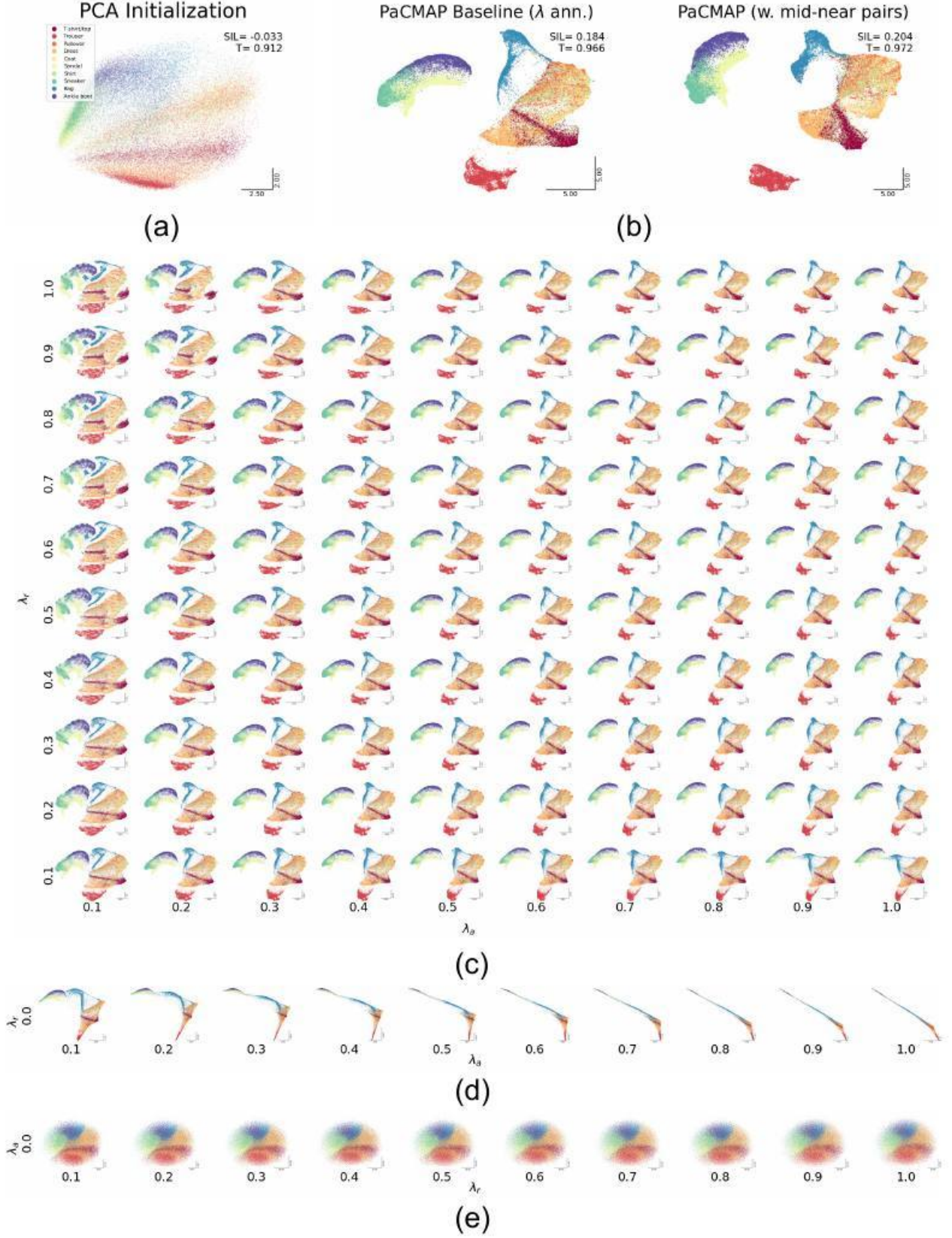


Figure 27: Learning-rate sweep over λ_a and λ_r on FMNIST using PaCMAP’s attraction and repulsion shapes. (a) Embedding initialization. (b) Left: baseline obtained by annealing λ from 1 with mid-near points excluded to isolate attraction–repulsion interaction; right: baseline with mid-near points included. (c) Embeddings obtained when λ_a and λ_r are varied without annealing. (d) When λ_r is set to 0 (no repulsion), attraction alone does not produce clusters. (e) Similarly, when λ_a is set to 0 (no attraction), repulsion alone does not produce clusters.

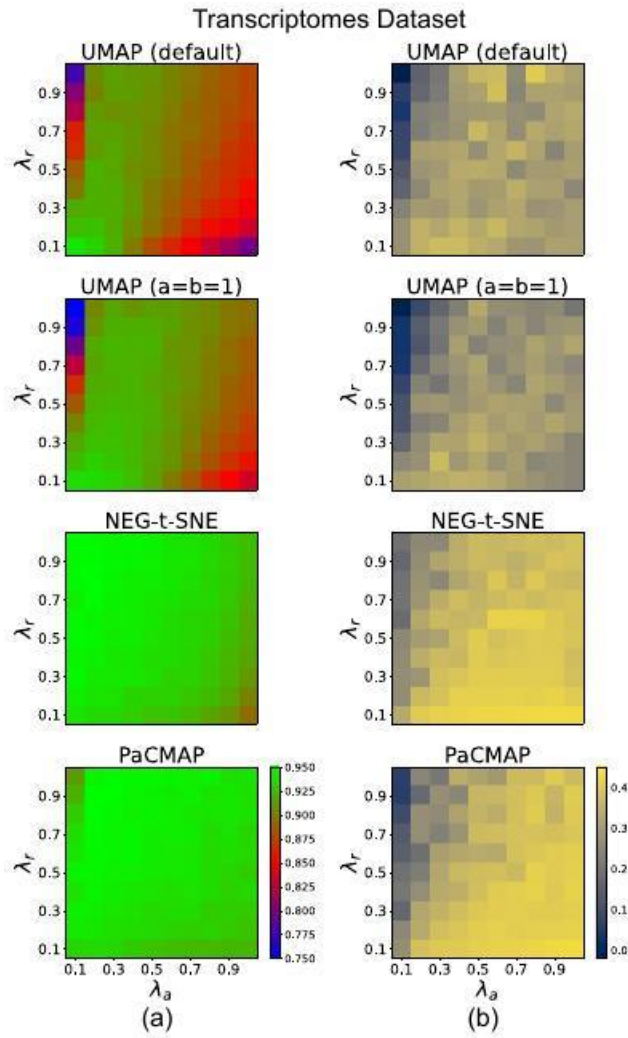


Figure 28: Single-cell-transcriptome evaluation metrics for the compared methods: (a) trustworthiness and (b) silhouette score.

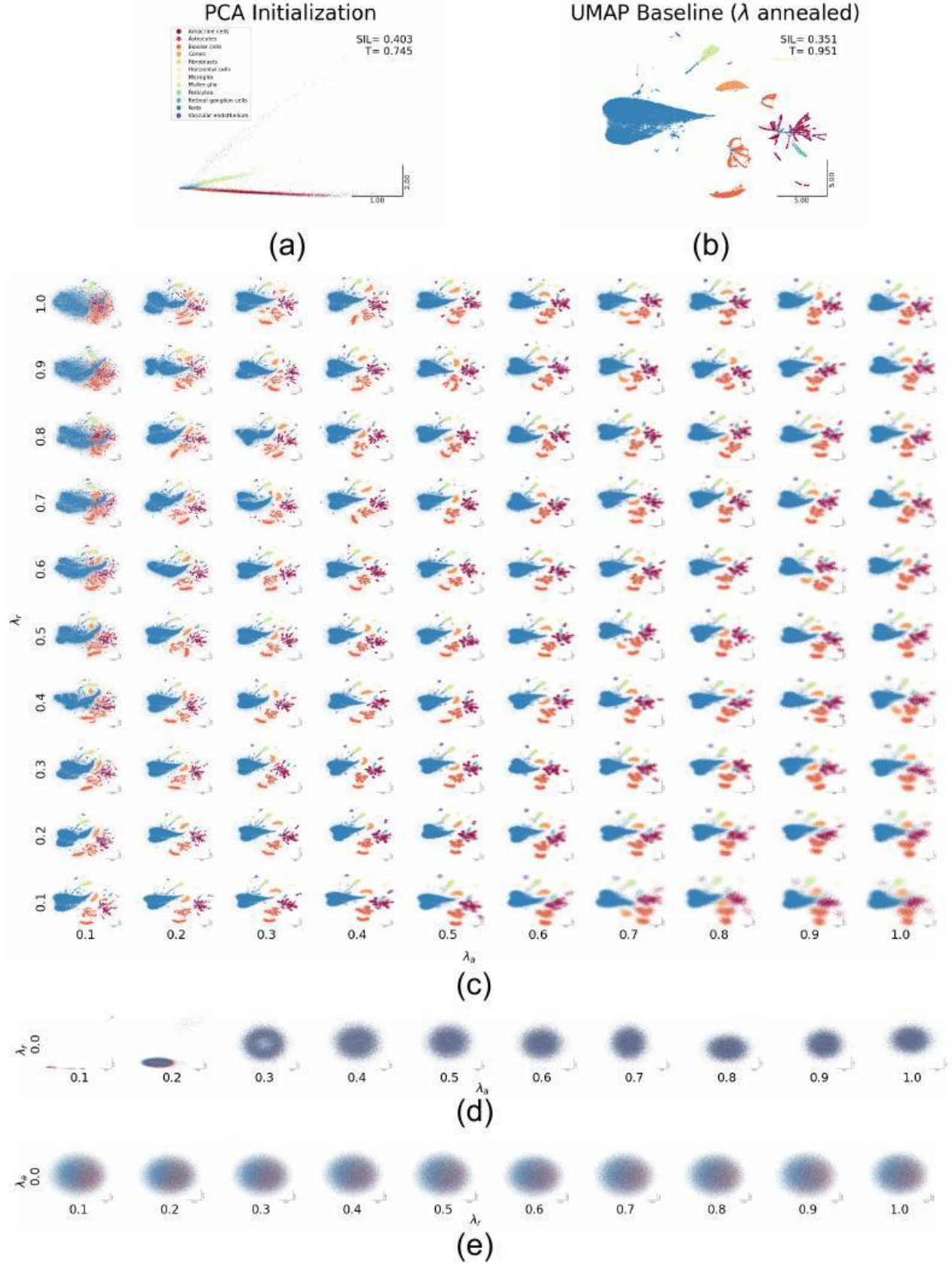


Figure 29: Learning-rate sweep over λ_a and λ_r on the single-cell transcriptomes dataset using UMAP's attraction and repulsion shapes. (a) Embedding initialization. (b) Baseline obtained by annealing λ from 1. (c) Embeddings obtained when λ_a and λ_r are varied without annealing. (d) When λ_r is set to 0 (no repulsion), attraction alone does not produce clusters. (e) Similarly, when λ_a is set to 0 (no attraction), repulsion alone does not produce clusters.

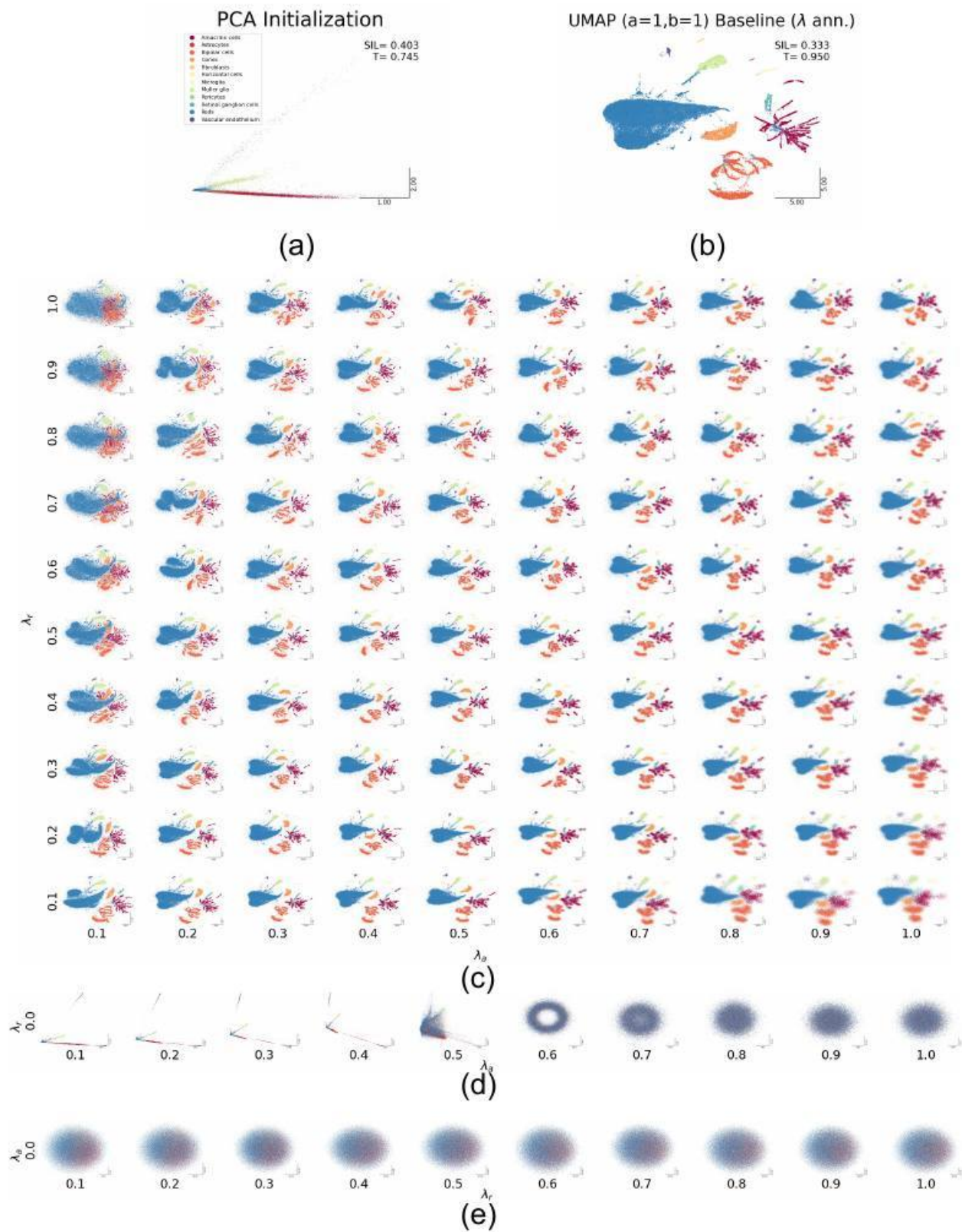


Figure 30: Learning-rate sweep over λ_a and λ_r on the single-cell transcriptomes dataset using UMAP’s attraction and repulsion shapes (by setting $a = 1$ and $b = 1$). (a) Embedding initialization. (b) Baseline obtained by annealing λ from 1. (c) Embeddings obtained when λ_a and λ_r are varied without annealing. (d) When λ_r is set to 0 (no repulsion), attraction alone does not produce clusters. (e) Similarly, when λ_a is set to 0 (no attraction), repulsion alone does not produce clusters.

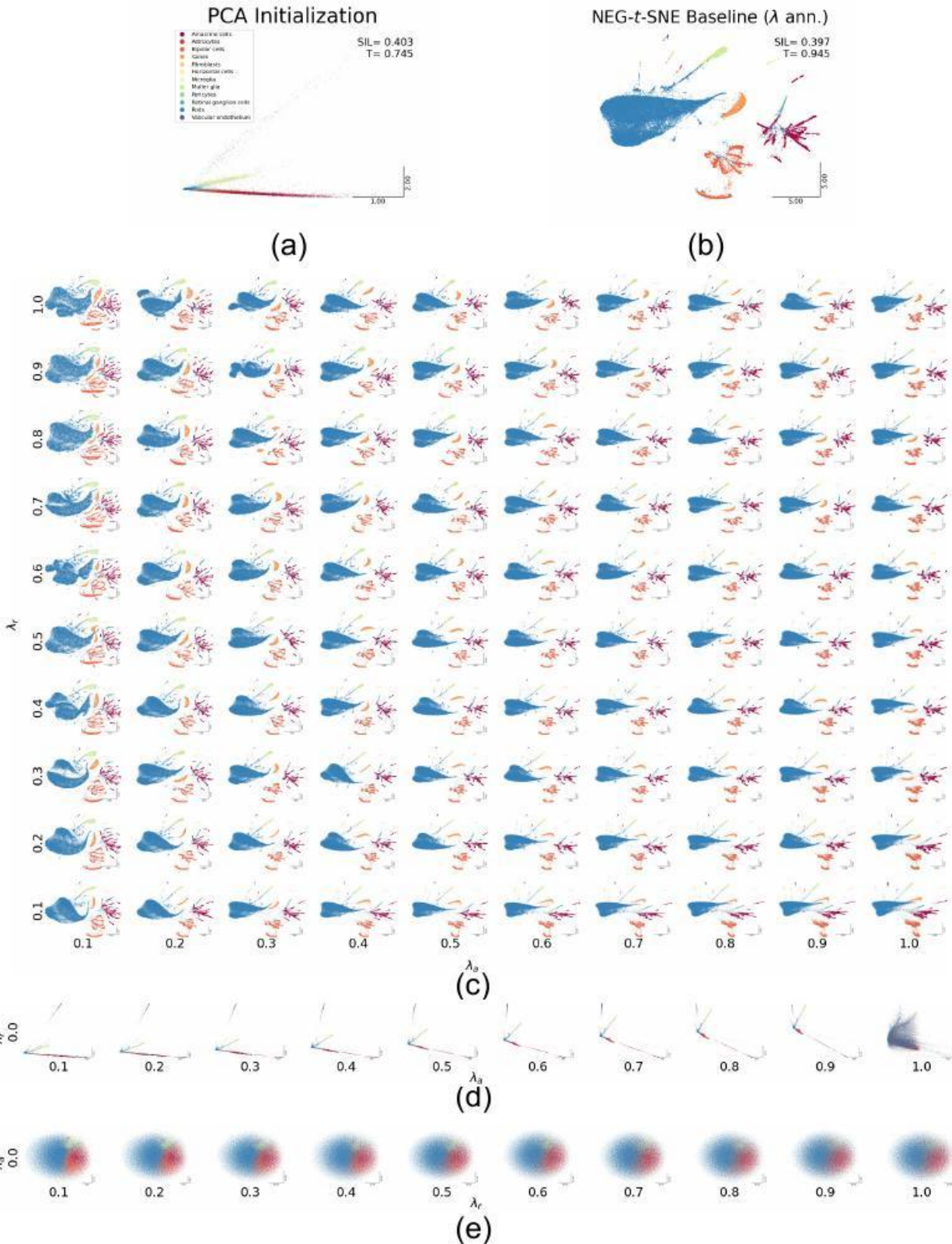


Figure 31: Learning-rate sweep over λ_a and λ_r on the single-cell transcriptomes dataset using NEG-*t*-SNE’s attraction and repulsion shapes. (a) Embedding initialization. (b) Baseline obtained by annealing λ from 1. (c) Embeddings obtained when λ_a and λ_r are varied without annealing. (d) When λ_r is set to 0 (no repulsion), attraction alone does not produce distinct clusters. (e) Similarly, when λ_a is set to 0 (no attraction), repulsion alone does not produce clusters.

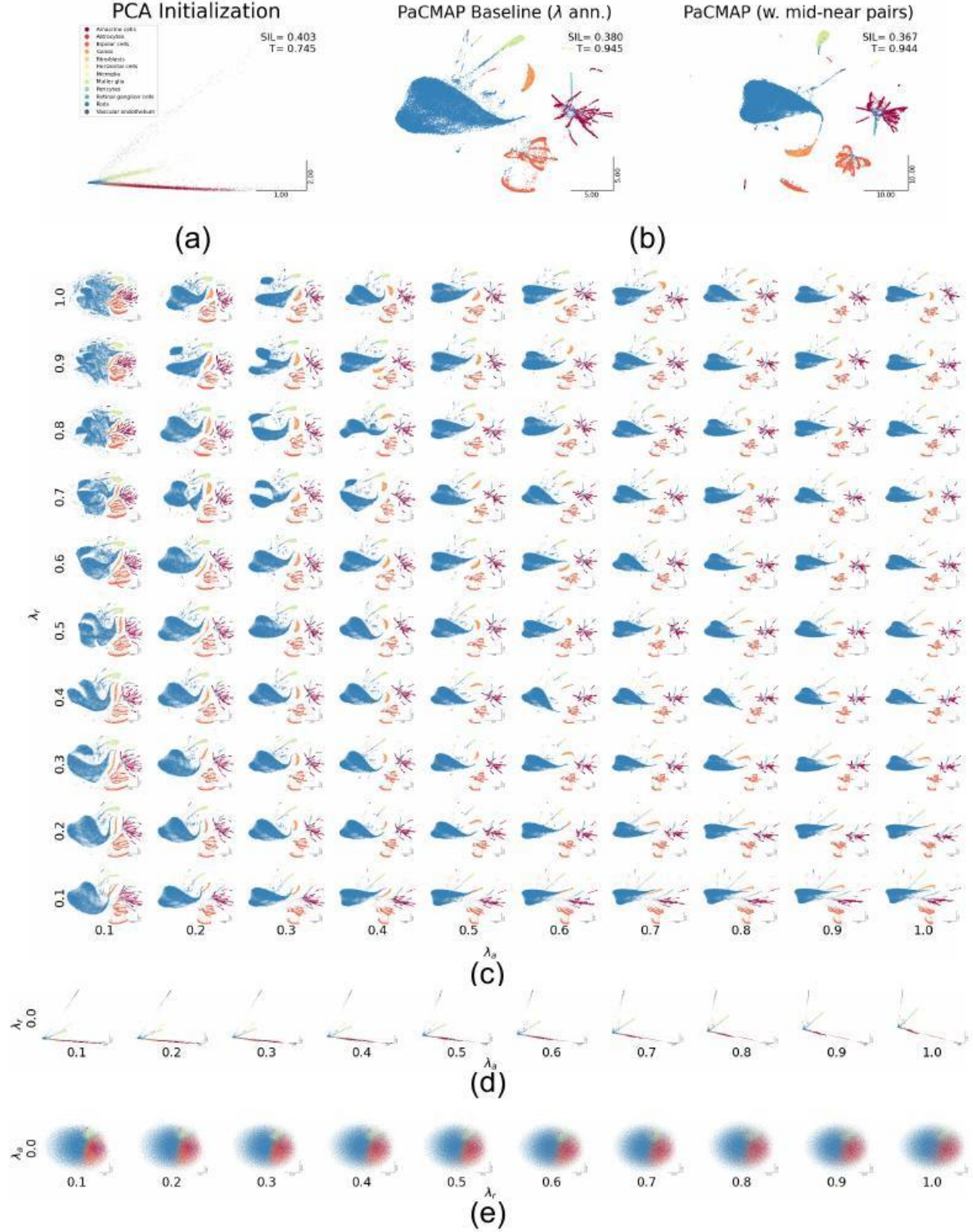


Figure 32: Learning-rate sweep over λ_a and λ_r on the single-cell transcriptomes dataset using PaCMAP's attraction and repulsion shapes. (a) Embedding initialization. (b) Left: baseline obtained by annealing λ from 1 with mid-near points excluded to isolate attraction–repulsion interaction; right: baseline with mid-near points included. (c) Embeddings obtained when λ_a and λ_r are varied without annealing. (d) When λ_r is set to 0 (no repulsion), attraction alone does not produce clusters. (e) Similarly, when λ_a is set to 0 (no attraction), repulsion alone does not produce clusters.

J Mix and Match Approach

The preceding analysis treats attraction and repulsion shapes as separable design objects. This perspective implies that shapes derived from different loss functions can be recombined. Figure 33 provides representative examples. In these experiments, attraction shapes are restricted to $[-1, 0]$ so that Proposition 4.1 applies and learning-rate sensitivity is reduced. Repulsion shapes are chosen to be finite, which limits excessively large repulsive updates at small distances. The learning rate is held fixed at $\lambda = 1$, and the same UMAP optimization procedure is used throughout, so differences are attributable to the shapes rather than to scheduling or optimizer changes. Overall, the scaled UMAP attraction shape ($f_a = 0.5f_a^U$) yields stronger clustering performance as measured by silhouette score, whereas the PaCMAP attraction shape yields better local-structure preservation as measured by trustworthiness. The latter effect is primarily due to PaCMAP’s attraction shape lying in $[-0.5, 0]$, which prevents pair flips during updates. Constraining the UMAP and NEG- t -SNE attraction shapes to $[-0.5, 0]$ via $f_a = 0.25f_a^U$ and $f_a = 0.5f_a^N$, respectively, improves local structure at the cost of clustering performance (Fig. 34).

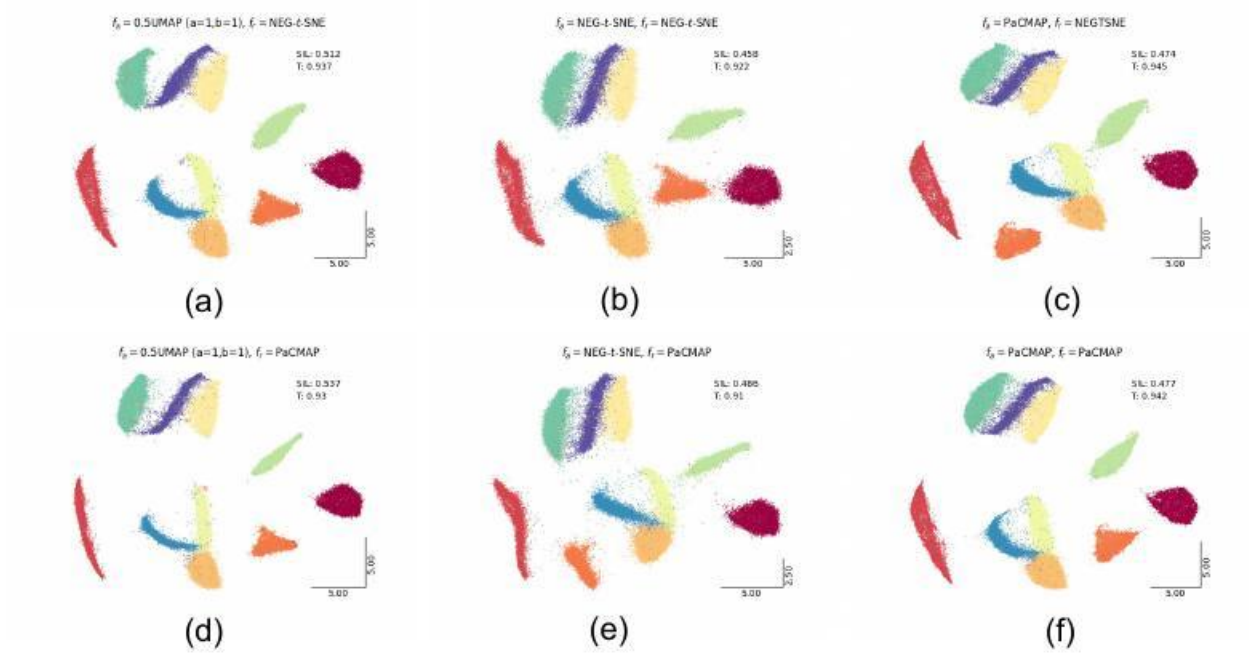


Figure 33: MNIST embeddings produced by cross-combining attraction and repulsion shapes at constant learning rate ($\lambda = 1$). Attraction shapes satisfy Proposition 4.1 and repulsion shapes are finite. Top row: UMAP, NEG- t -SNE, and PaCMAP attraction shapes paired with NEG- t -SNE repulsion. Bottom row: the same attraction shapes paired with PaCMAP repulsion.

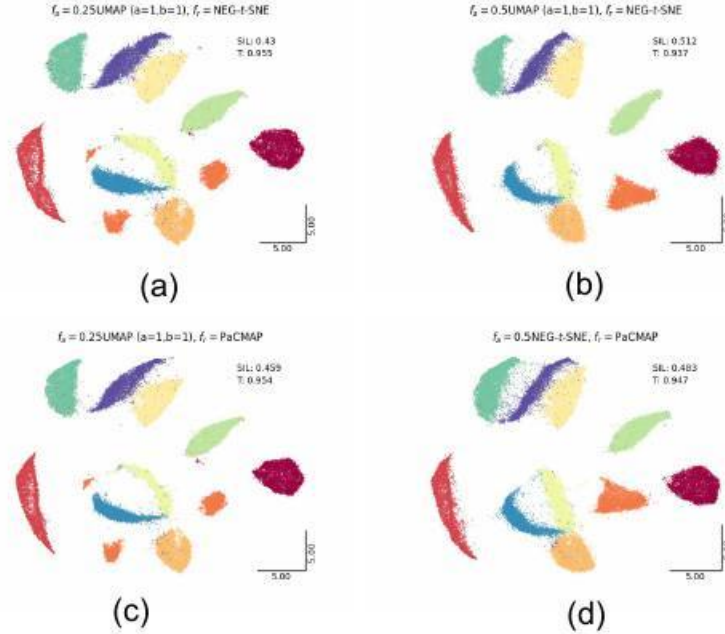


Figure 34: MNIST embeddings from cross-combined shapes at constant learning rate ($\lambda = 1$) with attraction confined to $[-0.5, 0]$ to prevent flips during attractive updates. Top row: UMAP and NEG- t -SNE attraction shapes paired with NEG- t -SNE repulsion. Bottom row: UMAP and NEG- t -SNE attraction shapes paired with PaCMAP repulsion.

K Implementation Details

For the analysis, we implemented a custom UMAP algorithm. We used numba (Lam et al. 2015) to compute an exact, rather than approximate, nearest-neighbor graph with $k = 15$, and we used the scikit-learn (Pedregosa et al. 2011) PCA implementation for PCA initialization. This implementation produced and quantified the embeddings in Figs. 1 2 3 4 6 8 9 and 10. The same implementation was used when substituting attraction and repulsion shapes from alternative methods (Figs. 18 34, unless otherwise stated). Trustworthiness and silhouette scores were computed with the corresponding scikit-learn functions.

The mappings in Figs. 2 3 9 10 and 15 are rotated to the reference embedding shown in panel (a) of the respective figure. We perform this alignment by normalizing embeddings to zero mean and unit norm, then applying SciPy's (Virtanen et al. 2020) `orthogonal_procrustes` method to estimate rotation and scaling parameters.

For the Neg- t -SNE comparison in Fig. 11, we used the original contrastive-embedding implementation for both UMAP and Neg- t -SNE; implementation details will be released upon acceptance.

PaCMAP and LocalMAP embeddings in Figs. 15 16 22 (b) (right), 27 (b) (right), and 32 (b) (right) were generated with the official PaCMAP package; implementation details will be released upon acceptance.

The code used to display the PubMed embeddings in Fig. 7 will be released upon acceptance.